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A 3-tier information fusioned framework featuring explainable deep active optimized CRNet for accurate heart disease prediction

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Abstract

Background Heart disease remains a leading cause of global morbidity and mortality, emphasizing the need for early and accurate prediction. Traditional models, however, often struggle with high class imbalance, limited accuracy, inadequate hyperparameter tuning, reliance on sparsely labeled data, and poor interpretability.

Methods To address these limitations, we introduce a novel 3-tier information fusioned framework integrating Oversampling Using the Propensity Score (OUPS), ConvRecurrentNet (CRNet), an optimized CRNet, and CRNet enhanced with Uncertainty-Based Sampling (UBS). Initially, the highly imbalanced heart disease health indicator dataset is balanced using OUPS while preserving the underlying data distribution. In the first tier, we develop a new deep model; CRNet which combines gated units to capture temporal dependencies and convolutional layers for efficient spatial feature extraction. The second tier leverages the grasshopper optimization algorithm to optimize CRNet configuration and classification performance through hyperparameter tuning. At the third tier, we use an active learning based UBS to address the problem of sparse labeled data selecting the most informative samples for effective CRNet model learning. Model performance and generalizability are assessed through 10-fold cross validation and one-way analysis of variance. Interpretability is ensured using explainable artificial intelligence (AI) methods, local interpretable model-agnostic explanations and shapley additive explanations.

Results Improvements are observed across all performance metrics: 3.45%, 5.75%, and 8.05% in accuracy, 3.45%, 6.90%, and 12.64% in precision, 4.65%, 5.81%, and 4.65% in recall, 4.65%, 6.98%, and 15.12% in F1-score, 1.04%, 2.08%, and 3.13% in receiver operating characteristic-area under the curve, 8.00%, 9.33%, and 17.33% in Matthews correlation coefficient, 12.16%, 13.51%, and 18.92% in Cohen's Kappa, and log loss reduction of 23.64%, 29.45%, and 45.82% across CRNet, optimized CRNet, and CRNet with UBS, respectively.

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Conclusion The proposed 3-tier fusion framework enhances heart disease prediction by balancing data, optimizing parameters, and applying active learning to address labeled data scarcity. Explainable AI further improves transparency and supports real-world clinical adoption.

Highlights

- OUPS balances HDHI dataset by reducing bias and preserving structure.
- Proposed CRNet, Optimized CRNet, and CRNet with UBS for prediction.
- 10-FCV with ANOVA ensures robustness and validates performance.
- LIME and SHAP enhance transparency and model interpretability.
- Generalizability tested on the heart failure prediction dataset.

Keywords Heart disease, Over-sampling using the propensity score, Grasshopper optimization algorithm, Uncertainty-based sampling, K-fold cross validation, One-way analysis of variance, Local interpretable model-agnostic explanation, Shapley additive explanations

Introduction

Healthcare significantly contributes to human well-being by preventing, identifying, and carefully treating illnesses that affect both individuals and communities. Healthcare has greatly improved its ability to identify and treat a wide range of medical conditions due to developments in medical technology and data-driven methodologies. But even with these developments, heart disease continues to be a major global public health concern. Life-threatening conditions can be lessened in impact by early diagnosis and risk assessment, which enables prompt interventions and better patient outcomes [1]. Heart disease continues to be the world's leading cause of death, with millions of deaths annually [2]. According to the World Health Organization, coronary heart disease accounts for nearly one-third of the approximately 17.9 million deaths caused by heart disease each year [3].

Heart disease, a major cause of morbidity and mortality, greatly impacts the world's healthcare systems. It includes heart failure, arrhythmias, valvular heart diseases, coronary artery disease, and other heart-related conditions [4]. Severe complications like heart attacks, strokes, and even sudden cardiac death can result from these conditions. Heart disease impacts quality of life, hospitalizations, and financial burdens in addition to personal health. It is still a top priority for medical research, preventative measures, and innovative treatment approaches because millions of cases are diagnosed each year [5].

Heart disease has severe consequences that affect physiological processes. It frequently causes decreased blood flow, which deprives essential organs of oxygen. As the heart muscle weakens over time, the heart finds it more difficult to pump blood effectively. Depending on the type and severity of the condition, heart disease symptoms can vary, but they typically include angina (chest pain), exhaustion, shortness of breath, irregular heartbeat, and swelling in the legs or feet. It often progresses silently over time, so early identification and risk factor evaluation are essential for efficient treatment [6].

Predictive models utilizing cutting-edge Machine Learning (ML), Deep Learning (DL), and Artificial Intelligence (AI) techniques have become more popular recently due to the potentially fatal nature of heart disease [7]. Healthcare practitioners can lower mortality rates, improve overall patient care, and accurately predict and detect heart disease early to provide tailored treatment plans. The incorporation of predictive analytics into healthcare creates new opportunities for identifying vulnerable people and putting preventative measures in place, which eventually improves public health outcomes [8]. The list of symbols and abbreviations is given in Table 1.

Related work

Numerous studies have explored the application of ML and DL techniques for heart disease prediction using a variety of datasets and feature selection techniques. To increase predictive accuracy, researchers have compared various computational methods for analyzing clinical and self-reported health indicators. Advanced frameworks use feature engineering, optimization, and regularization methods to improve model performance. Hybrid approaches and deep architectures are also the focus of recent research on finding intricate patterns in health data. This section provides previous research on computational methods for classifying heart disease.

The authors in [9] aim to determine the effectiveness of five ML models that can predict heart disease when self-reported physical health indicators are compared to all indicators checked by doctors. Logistic Regression (LR), K-Nearest Neighbors (KNN), Support Vector Model (SVM), Decision Tree (DT), and Random Forest (RF) are the five models used to forecast heart disease. The study's database includes 13 different types of test results and the heart disease risks for 303 patients. Six test results were available for the home matrices, while all matrices

Table 1 Acronyms used in the manuscript

Abbreviation	Description
ANN	Artificial Neural Network
BiLSTM	Bidirectional Long Short-Term Memory
CAD	Computer Aided Diagnosis
CNN	Convolutional Neural Network
DL	Deep Learning
ECG	Electrocardiogram
ESN	Echo State Network
GOA	Grasshopper Optimization Algorithm
GRU	Gated Recurrent Unit
K-FCV	K-Fold Cross Validation
LIME	Local Interpretable Model-agnostic Explanations
ML	Machine Learning
OUPS	Over-sampling Using the Propensity Score
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
SVM	Support Vector Machine
UBS	Uncertainty-Based Sampling
α	A random scalar drawn from a uniform distribution $U(0, 1)$
$b(l)$	Bias term at convolutional layer
h_t	Hidden state at time step t in a GRU cell
$\tilde{h}_t$	Candidate activation at time step t in a GRU cell
P	Set of prototype points selected from the minority class
$P(l)$	Output after applying the pooling operation at layer
p_i	The ith prototype in P
r_t	Reset gate at time step t in a GRU cell
$W(l)$	Convolution filter weights at layer
$W_o,\ b_o$	Weights and bias of the output layer
X	Feature set or dataset
x_i	The ith feature vector (sample) in the dataset
$X_{\min}$	Set of minority class samples in an imbalanced dataset
Y	Label set corresponding to X
y_i	The label corresponding to the ith sample
z_t	Update gate at time step t in a GRU cell
λ	Scaling parameter used in the OUPS method for synthetic sample generation
σ	Sigmoid activation function

included all thirteen test results. After constructing five models for both the home and all matrices, researchers calculated each model's accuracy score and False Negative (FN) rate. The findings demonstrated that in all five models, all matrices had greater accuracy scores than home matrices.

The authors in [10] use sophisticated neural network architectures to identify minute patterns and obscure relationships between different health indicators like blood pressure, cholesterol, and lifestyle choices. The model makes use of dense layers to model intricate patterns and forecast Cardiovascular Disease (CVD) risk, convolutional layers to extract spatial features, and embedding layers to convert categorical data into numerical representations. Regularization methods such as batch normalization and dropout, along with hyperparameter optimization, enhance the model's generalizability and performance. Extensive validation against traditional techniques shows that the model is superior, as evidenced by its noticeably higher R^2 value of 0.994.

Utilizing the patient's medical history, the authors in [11] developed a heart disease prediction system that determines the likelihood of a heart disease diagnosis. The authors classified and predicted the heart disease patients using various ML algorithms, including LR and KNN. The model applies a highly useful technique to enhance the accuracy of heart attack prediction in any individual. The proposed model's strength is fairly satisfying; it is able to use KNN and LR to predict evidence of heart disease in a specific individual with satisfactory

accuracy when compared to previously used classifiers like Naive Bayes (NB), etc.

The study's [12] purpose is to identify which ML classifiers achieve the highest accuracy. Several supervised ML methods are utilized, and their performance is compared. Ratings of estimated feature significance are assigned to each feature. Each feature is ranked according to its significance score. This study used a dataset on cardiac illness. A Kaggle dataset with multi-classes for classifying heart illness is used to train RF, DT, and KNN. All models achieved 100% accuracy, and RF provided 100% sensitivity and specificity.

To increase classification accuracy, the authors in [13] proposed a k-modes clustering technique with Huang's starting. RF, DT, Multilayer Perceptron (MLP), and eXtreme Gradient Boosting (XGB) are employed as base models. The applied model parameters are hypertuned using GridSearchCV to maximize the outcome. An actual dataset consisting of 70,000 instances from Kaggle is used to test the proposed model. DT, XGBoost, RF, and MLP achieved 86.3%, 86.87%, 87.05%, and 87.28% with cross-validation and 86.53%, 87.02%, 86.92%, and 86.94% without cross-validation accuracy. These models are trained on 80% of the data and tested on 20%. The MLP with cross-validation performed the best and outperformed baseline models.

ML techniques can be used to classify people into healthy and non-healthy categories. Using the clinical data parameters, the authors in [14] created a framework in this investigation that can comprehend the fundamentals of predicting a patient's risk profile. The proposed model is constructed using a χ^2-statistical model and a Deep Neural Network (DNN). On both the training and testing data, this model performs better and avoids overfitting and underfitting issues. The effectiveness of the model that correctly predicts the existence or absence of heart disease is examined using DNN and Artificial Neural Network (ANN).

To attain greater accuracy than conventional ML techniques, the authors in [15] proposed a hybrid model comprising Convolutional Neural Networks (CNN) with a Long Short-Term Memory (LSTM) network. A hybrid CNN and LSTM technique is used to classify the heart disease dataset as healthy and diseased. The K-FCV approach is utilized to confirm the 89% accuracy of this hybrid system. Proposed method is compared with ML algorithms such as SVM, NB, and DT to determine its effectiveness. The results show that the proposed algorithm outperforms the current ML models.

With the proposed approach, the authors use a set of images related to heart disease to construct the Levy Flight–CNN (LV-CNN) based on the diagnostic system for evaluating heart disease [16]. To lower the system's computational complexity, the input big data images are first resized. Thereafter, the proposed LV-CNN model is applied to those resized images. Consequently, the Sunflower Optimization Algorithm (SFO) and the LV approach are combined to lessen the loss function that arises in the CNN architecture. This combination helps the SFO algorithm stay out of local minima caused by the LVs' random walk.

In [17], the authors use CNN and Bi-directional LSTM (Bi-LSTM) to build a hybrid model based on DL techniques for determining if a person has heart disease and then using that prediction to provide awareness or a diagnosis. Using data processing techniques, the authors resolve the missing data and imbalanced data issues in the publicly available Cleveland dataset gathered from the University of California, Irvine, for heart disease. The authors employed CNN-BiLSTM for classification and an additional tree classifier for feature selection.

To reduce the drawbacks of the traditional ML techniques, the authors in [18] suggest a novel DL architecture that uses a 1D CNN for classification between healthy and non-healthy individuals with balanced datasets. The risk contour of the patient is assessed using several clinical parameters that aid in early diagnosis. To prevent overfitting in the proposed model, a variety of regularization techniques are applied.

The existing studies demonstrate how computational methods for predicting heart disease are constantly improving, spanning from conventional methods to DL-based frameworks. Existing research incorporates optimization techniques, hybrid architectures, and feature selection strategies to improve accuracy and reliability, whereas existing methods concentrated on basic predictive modeling. Data imbalance, generalizability, and interpretability are still major issues despite the previous advancements. The summary of related work is given in Table 2.

Research gaps in existing studies, salient contributions and paper organization

This section discusses the research gaps, contributions, and organization of the paper.

Research gaps in existing studies

Heart disease detection has advanced significantly, but several key challenges still limit the effectiveness and reliability of predictive models. According to the related work, the current ML and DL models suffer from the following research gaps:

- Data Imbalance Handling: When dealing with class imbalance, existing models finds it difficult to preserve the original data distribution. These methods introduce noise or the inability to identify underlying data patterns that can result in biased

Table 2 Comparison of existing methods and proposed solution

Methods	Methodology	Limitations	Proposed Solution
LR, KNN, SVM, DT, RF [9]	Predict heart disease using self-reported versus doctor-checked indicators on 303 patients (home matrices with six tests vs full matrices with thirteen tests)	Limited dataset, reliance on self-reported data, lower accuracy for home matrices	Proposed CRNet with comprehensive clinical data and data balancing to improve accuracy and robustness
Neural Networks (Dense, CNN, Embedding layers) [10]	Capture intricate interactions among health indicators; extract spatial features; convert categorical data via embeddings; regularized with batch normalization, dropout, and hyperparameter tuning (yielding high R^2)	High computational complexity; potential lack of model interpretability.	Integrate explainable AI (Local Interpretable Model-agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP)) with CRNet to maintain performance while enhancing interpretability
LR, KNN [11]	Utilize patient medical history to predict the likelihood of heart disease with traditional ML classifiers	Limited to conventional ML; cannot generalize well across diverse patient profiles.	Employ CRNet that enhances feature extraction and generalization across varied datasets
RF, DT, KNN [12]	Compare various ML classifiers on a three-class Kaggle dataset; rank features by significance; some models report 100% accuracy	Unrealistically high accuracy can indicate overfitting and poor generalizability.	Adopt robust cross-validation (e.g., 10-FCV) and CRNet to ensure realistic performance and avoid overfitting
DNN, χ^2-statistical model, ANN [14]	Build a framework using clinical data parameters to predict a patient's risk profile; addresses overfitting/underfitting via DL models	Can lack advanced feature extraction and interpretability compared to more complex architectures.	Integrate convolutional and recurrent components in CRNet along with explainable AI methods for improved predictive insight
Hybrid CNN + LSTM [15]	Classify heart disease as healthy/heart disease using a deep hybrid model validated via K-FCV (accuracy around 89%)	Training complexity and moderate accuracy; limited explainability	Propose CRNet that combines gated recurrent units with convolutional layers for higher accuracy and enhanced transparency
LV-CNN with SFO [16]	Use resized heart disease image data processed by an LV-CNN; combine SFO and LV to lessen the loss function and avoid local minima	Focuses solely on image data; computationally intensive and complex to optimize	Focus on structured clinical data with CRNet to achieve robust predictions with lower computational cost

predictions. More efficient balancing strategies that enhance minority class representation without sacrificing data integrity are required.

- Feature Integration and Extraction: Although models that combine various feature extraction mechanisms have demonstrated promise, they still haven't reached their full potential. Maximizing the integration of spatial and sequential features can improve the model's ability to detect important patterns in intricate data.
- Hyperparameter Optimization Techniques: Selecting the right model parameters significantly impacts a predictive model's performance. Traditional approaches for parameter tuning often fail to find the optimal configurations and can be computationally costly. To attain the best model performance, more effective and perceptive search techniques are needed.
- Active Learning Techniques: Performance can be greatly enhanced during the model training process by choosing informative samples, particularly in situations where there is a shortage of labeled data. Nevertheless, there is an opportunity to improve learning efficiency because the use of approaches that assign priority to ambiguous or uncertain samples is not extensively investigated.
- Model Interpretability and Explainability: Since many models operate as black boxes, it can be challenging to comprehend how predictions are made. Explainable AI techniques are required to offer insights into feature contributions and model decisions.

Salient contributions (addressing the research gaps)

To address the research gaps mentioned in Sect. 3.1, the present study suggests a comprehensive framework that utilizes cutting-edge AI techniques to enhance heart disease prediction.

- The Over-sampling Using the Propensity Score (OUPS) technique is used to overcome the inherent imbalance in the Heart Disease Health Indicator (HDHI) dataset. To create synthetic samples, OUPS balances the distribution of covariates, minimizes selection bias, and maintains the underlying structure of the dataset using the propensity score.
- We proposed a 3-Tier Information Fusioned Framework (3TIFF) that fuses the information coming from OUPS, ConvRecurrentNet (CRNet), Optimized CRNet, and CRNet with Uncertainty-Based Sampling (UBS) to predict heart disease at early stages. In the first tier, we develop a novel CRNet model that combines gated units to capture temporal dependencies and convolutional layers to extract spatial features in the dataset. In the second tier, the CRNet model's hyperparameters are optimized using the Grasshopper Optimization Algorithm (GOA). GOA successfully looks for

ideal configurations by striking a balance between exploration and exploitation, inspired by the swarming behavior of grasshoppers. In the third tier, an active learning technique called UBS is used with the CRNet model. It methodically chooses the most ambiguous samples, those nearest to the decision boundary, to ensure that the model trains from the most informative instances.

- 10-Fold Cross Validation (10-FCV) is utilized to evaluate the model's performance and generalizability, ensuring that performance metrics remain consistent across various data splits. This extensive assessment validates the model's robustness and reliability, lowering the risk of overfitting. After applying 10-FCV, the model's performance differences are also statistically analyzed using One-Way Analysis of Variance (ANOVA), which validates the significance of the observed variations across various evaluation metrics.
- To enhance the interpretability and explainability of the CRNet model, explainable AI techniques such as Local Interpretable Model-agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP) are employed. LIME offers interpretability by revealing the decision-making process of CRNet for every prediction, while SHAP explains model predictions according to the contribution of specific features globally.
- To further assess generalizability, we also test our proposed models on the heart failure prediction dataset, which is a combination of five benchmark datasets.

Paper organization

The paper is structured as follows: Sect. 1 represents the brief introduction of the healthcare domain. The relevant work is presented in Sect. 2. Sect. 3 highlights the research gaps, contributions, and paper organization. The discussion of the proposed system models is discussed in Sect. 4. Section 5 evaluates the simulation-based results carried out using the proposed methodology. Section 6 wraps up the study and addresses potential avenues for future research.

System model

This section provides an in-depth discussion of the proposed CRNet model's process. During the preparation phase, duplicate instances are removed to ensure data consistency. Thereafter, the feature values in the dataset are then standardized by using the standard scaler. Next, the OUPS oversampling technique, which generates synthetic samples for the minority class, is proposed to address class imbalance and reduce bias. Later, 3TIFF is proposed for early heart disease prediction, and its working is illustrated in Fig. 1. At the first tier, a novel CRNet model is developed, a combination of gated units and convolutional layers. It improves predictive performance by utilizing both sequential dependencies and spatial feature extraction. In the second tier, the CRNet model undergoes additional optimization with GOA to fine-tune hyperparameters for enhanced performance. Finally, in the third tier, the most informative samples are chosen from unlabeled data, and the UBS active learning technique is used, which minimizes annotation effort and enhances model generalization. The methodology used in this study is described in Table 3. The flow diagram of 3TIFF is shown in Fig. 2.

Description of heart disease health indicator dataset

The Behavioral Risk Factor Surveillance System (BRFSS) is a health-related telephone survey that is collected annually by the Center of Disease Control (CDC). Each year, the survey collects responses from over 400,000 Americans on health-related risk behaviors, chronic health conditions, and the use of preventative services. Since 1984, the CDC has conducted it annually. This HDHI dataset contains 253,680 survey responses from the cleaned BRFSS 2015 to be used primarily for the binary classification of heart disease. The HDHI dataset is nationally representative, covering a diverse population across different ages, genders, ethnicities, and geographic regions in the United States (U.S.). Its large and diverse sample ensures that the dataset can reliably reflect the broader U.S. adult population, making it suitable for modeling and predicting heart disease risk.

The HDHI dataset is used for predicting heart disease that consists of 253,681 instances and 21 features [19]. The dataset represents an imbalanced classification problem: Class 1 (with 23,888 instances or 0.87%) corresponds to people diagnosed with heart disease, while Class 0 (with 229,773 instances or 94.13%) represents healthy individuals. The dataset includes a mix of self-reported health conditions (*general health rating*, *mental health*, and *physical health*), lifestyle choices (such as *smoking*, *drinking*, and *physical activity*), and medical history (such as *diabetes*, *high blood pressure*, and *high cholesterol*). To provide a more comprehensive understanding of health disparities, the dataset also includes socioeconomic factors like *income*, *education level*, and access to healthcare services. We used only 50% of the dataset to obtain effective results. A complete description of the dataset is provided in Table 4.

The dataset contains duplicate rows to eliminate them, we use Python's *drop_duplicates()* function [20]. This function identifies and eliminates repeated rows, ensuring that each record in the dataset is unique. Mathematically, duplicate detection can be represented as:

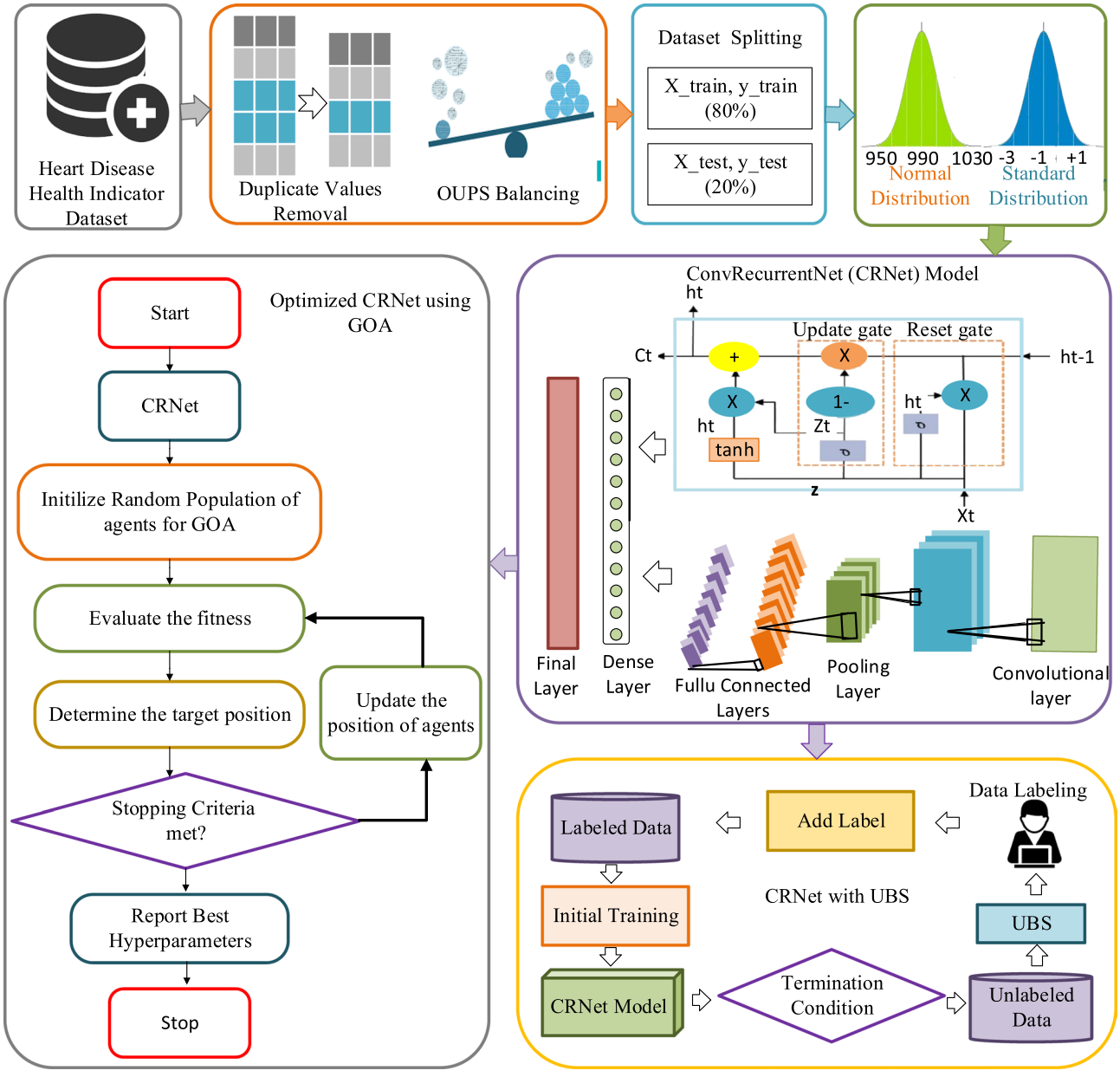


Fig. 1 Newly proposed 3-tier information fusioned framework (3TIFF) for accurate heart disease prediction at early stages

$$D' = D - \{x_i \mid x_i = x_j, \forall i \neq j\} \quad (1)$$

where D is the original dataset, D' is the cleaned dataset, and x_i, x_j are duplicate instances in D. Removing duplicates helps improve the quality of training data and prevents biased learning.

Data balancing with over-sampling using the propensity score

Class imbalance occurs when one class in a dataset has substantially more samples than the other and is addressed through balancing in ML and DL. This imbalance leads to biased model predictions that favor the majority class, making it harder for the model to accurately classify minority instances. Conventional methods of balancing, like random oversampling and undersampling, frequently entail the occurrence of noise or the lack of crucial information. Techniques like Synthetic Minority Over-sampling (SMOTE) generate synthetic samples, but they do not always preserve the underlying distribution of the data [21]. Class imbalance occurs when one class in a dataset has substantially more samples than the other and is addressed through balancing in ML and DL. This imbalance leads to biased model predictions that favor the majority class, making it harder for the model to accurately classify minority instances. Conventional

Table 3 Summary of methodology

Method	Description	Key Features	Applications
Standard Scaling	A normalization technique that scales features to a standard normal distribution.	Transforms data to zero mean and unit variance.	Used for preprocessing before applying DL models.
OUPS Balancing	A method for balancing datasets by modifying underrepresented classes.	Adjusts the data distribution to ensure all classes have equal importance.	Applied in classification tasks to handle imbalanced datasets.
CRNet	A DL model consisting of convolutional and a gated module.	Includes convolutional and pooling layers and a gating mechanism to improve learning and performance.	Applied in various DL tasks such as classification and regression.
Optimized CRNet	CRNet model optimized using the GOA.	Improved hyperparameter tuning for better performance and generalization.	Applied in binary classification tasks with enhanced predictive accuracy.
CRNet with UBS	CRNet model integrated with UBS for active learning.	Selects the most uncertain samples for annotation to improve model performance.	Applied in scenarios with limited labeled data to enhance model efficiency.
K-FCV	A model validation technique that splits the data into K subsets.	Involves training the model K times, each with a different subset as the test set.	Used to assess the performance and generalization of models.
SHAP	A method for explaining DL models by calculating Shapley values.	Provides feature importance and explains model predictions.	Used for model interpretability and feature attribution.
LIME	A technique for explaining model predictions locally by approximating the model with simpler interpretable models.	Creates locally interpretable surrogate models to explain individual predictions.	Applied in scenarios requiring interpretable AI and model transparency.

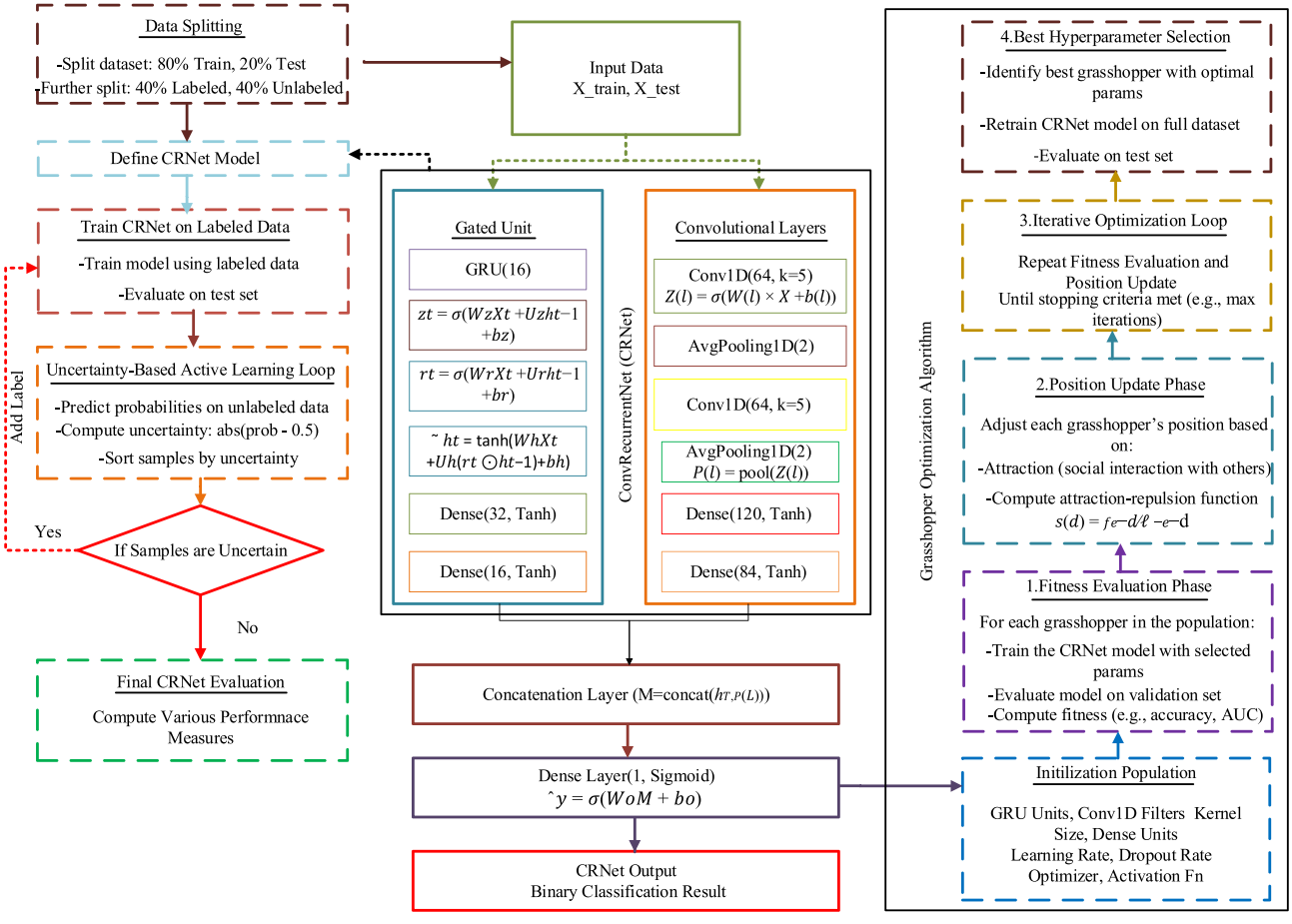


Fig. 2 Flow diagram of 3-tier information fusioned framework (3TIFF) for accurate heart disease prediction at early stages

Table 4 Heart disease health indicator dataset description

Attribute	Values	Description
HeartDiseaseorAttack	0 = No, 1 = Yes	Indicates if the person has a history of heart disease or heart attack.
HighBP	0 = No, 1 = Yes	Diagnosed with High Blood Pressure.
HighChol	0 = No, 1 = Yes	Diagnosed with High Blood Cholesterol.
CholCheck	0 = No, 1 = Yes	Cholesterol checked within the last 5 years.
BMI	16–59	Body Mass Index, calculated by dividing weight (kg) by the square of height (m).
Smoker	0 = No, 1 = Yes	Smoked at least 100 cigarettes in their lifetime.
Stroke	0 = No, 1 = Yes	History of stroke.
Diabetes	0 = No, 1 = Yes, 2 = Borderline	History of diabetes, pre-diabetes, or current diabetes diagnosis.
PhysActivity	0 = No, 1 = Yes	Engages in physical activity in daily routine.
Fruits	0 = No, 1 = Yes	Consumes 1 or more fruit daily.
Veggies	0 = No, 1 = Yes	Consumes 1 or more vegetables daily.
HvyAlcoholConsump	0 = No, 1 = Yes	Drinks more than 14 alcoholic beverages per week.
AnyHealthcare	0 = No, 1 = Yes	Has health insurance.
NoDocbcCost	0 = No, 1 = Yes	Wanted to see a doctor in the past year but couldn't due to cost.
GenHlth	Scale: 1 (Excellent) to 5 (Poor)	Self-reported general health condition.
MentHlth	Number of days (0–30) with poor mental health	Number of days in the past 30 days with bad mental health.
PhysHlth	Number of days (0–30) with poor physical health	Number of days in the past 30 days with bad physical health.
DiffWalk	0 = No, 1 = Yes	Difficulty walking or climbing stairs.
Sex	0 = Female, 1 = Male	Gender of the person.
Age	Categorical: 1 (18–24), 2 (25–29), … , 13 (80+)	Age group, with each interval representing a 5-year increment.
Education	Categorical: 1 (Never attended school) to 6 (College graduate)	Highest level of education completed.
Income	Categorical: 1 (Less than $10,000) to 8 ($75,000 or more)	Total household income level.

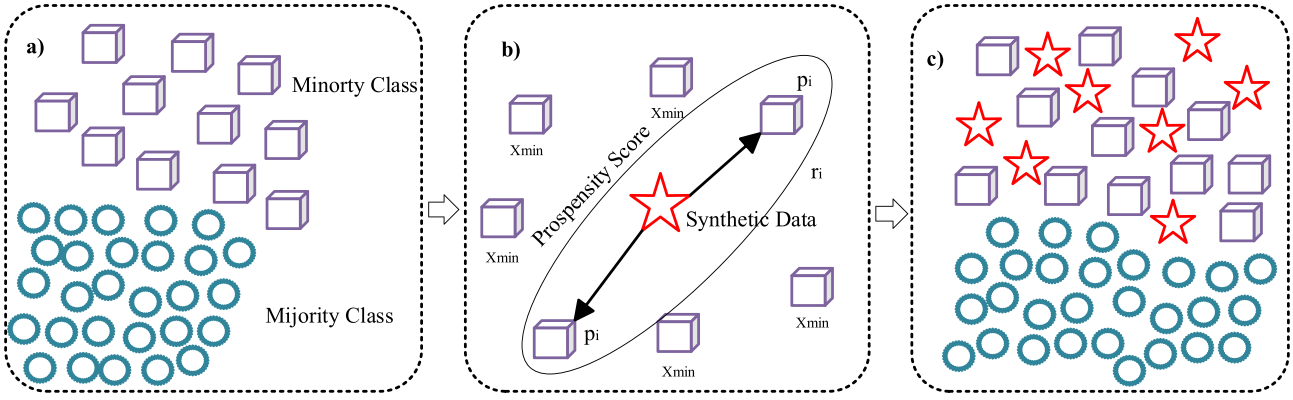


Fig. 3 Working of over-sampling using the propensity score

methods of balancing, like random oversampling and undersampling, frequently entail the occurrence of noise or the lack of crucial information. These techniques fail to consider the uncertainty of samples, a crucial factor in classification tasks.

The OUPS method overcomes these limitations by utilizing propensity scores, which measure the probability of an event occurring given a set of observed covariates [22]. Strategic sample selection by OUPS is based on classification uncertainty rather than random oversampling or undersampling. This is achieved by calculating each instance's propensity score, which indicates that, based on its features, it is a part of the minority class. Samples with a lot of uncertainty are either matched based on propensity scores or used as extra information in the feature space if the model isn't sure how to classify them [23]. The resampled dataset is guaranteed to maintain its original distribution, and the model's sensitivity to minority class instances is enhanced. In contrast to conventional techniques, OUPS introduces uncertainty into the sampling procedure, producing a model that is more impartial and reliable.

The working of OUPS is illustrated in Fig. 3. The initial class imbalance in which the majority class greatly

outnumbers the minority class is depicted in Fig. 3(a). The model's capacity to accurately classify instances from the minority class is diminished as a result of this imbalance, which biases them toward the majority class. A set of prototype points P is selected from the minority class using a clustering-based approach:

$$P = \{p_1, p_2, ..., p_m\}, \quad P \subset X_{min} \tag{2}$$

where X_{min} represents the samples of the minority class. For each prototype p_i, a sphere is constructed with radius r_i, defined as:

$$r_i = \lambda \cdot d_{\text{avg}}(p_i, X_{\text{min}}) \tag{3}$$

where $d_{\text{avg}}(p_i, X_{\text{min}})$ is the average distance from the prototype p_i to its nearest minority class neighbors, and λ is a scaling parameter (set to the default value of 1.0 in our experiments). Figure 3(b) shows this process. Finally, synthetic samples X_{syn} are generated within each sphere by randomly interpolating between the prototype and its neighboring samples:

$$x_{\text{new}} = p_i + \alpha \cdot (x_{\text{neighbor}} - p_i) \tag{4}$$

where $x_{\text{neighbor}} \in X_{\text{min}}$ is a randomly chosen neighboring sample, and $\alpha \sim U(0, 1)$ is a random scalar drawn from a uniform distribution:

$$X_{\text{resampled}} = X \cup X_{\text{syn}}, \quad Y_{\text{resampled}} = Y \cup Y_{\text{syn}} \tag{5}$$

where X_{syn} and Y_{syn} represent the synthetic samples and their corresponding labels. The oversampled data after applying OUPS is illustrated in Fig. 3(c).

OUPS provides a more efficient method of addressing class imbalance, improving predictive model robustness and generalization. This benefit is especially advantageous in predicting heart disease and the larger healthcare field, where accurately identifying minority class instances can improve early diagnosis and focused interventions.

Data splitting

For conventional model training, the resampled dataset is divided into training (80%) and testing (20%) sets. For the active learning setup, the training set is further split into labeled (40%) and unlabeled (40%) subsets to simulate limited labeled data availability, with a random seed of 42 set for reproducibility.

Scaling improves model performance and generalization [24], reduces the effect of absolute feature values and highlights the relative importance of features. After splitting the dataset into training and testing sets, standard scaling is applied [25]. This transforms the features to a mean of 0 and a standard deviation of 1, ensuring all features have a standard normal distribution:

$$X' = \frac{X - \mu}{\sigma} \tag{6}$$

where X is the original feature value, μ is the mean of the feature, σ is the standard deviation, and X' is the standardized value. This transformation ensures that all features have comparable scales, improving convergence in gradient-based optimization algorithms.

Proposed ConvRecurrentNet deep model for accurate heart disease prediction at early stages

Through the integration of various convolutional, pooling, and gated units, the CRNet model is intended to effectively capture both spatial and sequential patterns in data. The model is well-suited for challenging classification tasks because of its architecture, which allows it to efficiently learn hierarchical features while maintaining temporal dependencies. CRNet's capacity to extract significant spatial features via its convolutional layers is one of its main advantages. These layers gradually construct higher-level representations and assist in identifying local patterns like edges and textures. The model improves computational efficiency and generalization by reducing dimensionality while preserving the most crucial information when combined with pooling operations. Apart from its robust feature extraction capabilities, CRNet integrates specialized gated mechanisms that improve its capacity to identify inter-sequence dependencies. These gated units regulate the information flow by removing less helpful aspects and retaining just those that are relevant. This keeps crucial contextual information from being lost over time by enabling the model to preserve long-range dependencies. For structured data where both spatial and sequential characteristics are essential, CRNet's combination of convolutional and gated components makes it especially effective. It performs better on tasks requiring a thorough comprehension of spatial patterns and temporal relationships by utilizing the strengths of these various components. The CRNet model captures both spatial and sequential dependencies in data by integrating convolutional and gated recurrent components. Given an input sequence $X \in \mathbb{R}^{T \times d}$, where T represents the time steps and d is the feature dimension, the model processes data through convolutional and gated mechanisms.

Both components in the architecture are designed to run in parallel. The same input feature vector, denoted as X_{train}, is independently supplied to the gated component and to the convolutional component. Each component performs a distinct type of processing: the gated component emphasizes the extraction of sequential

dependencies across the features, while the convolutional component applies one-dimensional convolutions to capture local interactions and structural patterns among neighboring features. Once the two components have completed their respective computations, their outputs are concatenated at the final stage. This fusion enables the model to integrate sequential information with localized feature representations, thereby enhancing the robustness and accuracy of the final prediction.

The working of CRNet is mentioned in Algorithm 1. The algorithm begins by defining the dataset X, which contains relevant health features such as *HeartDiseaseAttack*, *BMI*, *HighBP*, *HighChol*, *Smoker*, *Stroke*, *Diabetes*, *Fruits*, *GenHlth*, *MentHlth*, *Sex*, and *Age*, among others. The primary objective is to predict the probability $\hat{y}$ of heart disease occurrence based on these features. The convolutional component extracts spatial features by applying a set of filters as mentioned in Algorithm 1 (lines 3–7):

$$Z^{(l)} = \sigma(W^{(l)} \times X + b^{(l)}) \tag{7}$$

where $W^{(l)}$ represents the convolutional filter weights, $\times$ denotes the convolution operation, $b^{(l)}$ is the bias term, and σ is the activation function. To reduce dimensionality while preserving important information, average pooling is applied:

$$P^{(l)} = \text{pool}(Z^{(l)}) \tag{8}$$

where $\text{pool}(\cdot)$ represents the average pooling operation. In parallel, the model processes sequential dependencies using gated units. The update and reset gates are computed, as given in Algorithm 1 (lines 8–14):

$$z_t = \sigma(W_z X_t + U_z h_{t-1} + b_z) \tag{9}$$

$$r_t = \sigma(W_r X_t + U_r h_{t-1} + b_r) \tag{10}$$

where z_t and r_t are the update and reset gates, respectively, W_z, W_r, U_z, U_r are learnable weight matrices, and b_z, b_r are biases. The candidate activation is computed as:

Table 5 Hyperparameters utilized in CRNet

Hyperparameter	CRNet
Gated Units	16
Dense Units	32, 16
Filters	64
Kernel Size	5
Activation Function	Sigmoid, Tanh
Pooling Size	2
Dense Units	120, 84
Optimizer	Adam
Batch Size	64
Epochs	10

$$\tilde{h}_t = \tanh(W_h X_t + U_h(r_t \odot h_{t-1}) + b_h) \tag{11}$$

where the term $\tilde{h}_t$ represents the candidate hidden state at the time step t, which is computed based on the current input and the previous hidden state. The function $\tanh(\cdot)$ applies a non-linearity to regulate the activation values. The term $W_h X_t$ captures the transformation of the current input X_t using the weight matrix W_h. The component $U_h(r_t \odot h_{t-1})$ incorporates the previous hidden state h_{t-1}, modulated by the reset gate r_t, where U_h is the corresponding weight matrix. Finally, b_h represents the bias term, which helps adjust the learned transformation. The final hidden state update is given by:

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \tag{12}$$

where $\odot$ represents element-wise multiplication. The outputs from the convolutional and gated branches are then concatenated as demonstrated in Algorithm 1 (line 15):

$$M = \text{concat}(h_T, P^{(L)}) \tag{13}$$

where h_T is the final hidden state and $P^{(L)}$ represents the final pooled convolutional features. Then the combined representation M is passed through fully connected layers as mentioned in Algorithm 1 (line 16):

$$y = \sigma(W_o M + b_o) \tag{14}$$

where W_o and b_o are the weights and bias of the output layer, and σ is the *Sigmoid* activation function, yielding a probability score for binary classification. A list of the hyperparameters utilized in the CRNet model is shown in Table 5. The CRNet model uses several health indicators to predict the likelihood of heart disease. The hyperparameters include the number of gated units, dense layers, convolutional layers, filters, kernel size, activation functions, pooling layer, optimizer, batch size, and epochs. The model makes use of two dense layers with 32 and 16 units each, as well as 16 gated units. To extract spatial patterns from the data, 64 filters with a kernel size of 5 are applied to the convolutional layers. By using the *Sigmoid* and *Hyperbolic Tangent (Tanh)* activation functions, the learning process is guaranteed to be non-linear. By downsampling the feature maps using a pool size of two, the pooling layers lower computational complexity. The *Adaptive Moment Estimation (Adam)* optimizer, which has a batch size of 64 and trains over 10 epochs, is chosen for its adaptive learning rate capabilities. These hyperparameters are crucial for assessing the model's performance because they allow it to accurately predict heart disease by capturing both spatial and temporal dependencies.

Algorithm 1 ConvRecurrentNet for Accurate Heart Disease Prediction at Early Stages

Input: Dataset X with features {HeartDiseaseAttack, BMI, HighBP, HighChol, Smoker, Stroke, Diabetes, Fruits, GenHlth, MentHlth, Sex, Age, etc.}
Output: Predicted probability of heart disease $\hat{y}$
Initialize convolutional filters $W^{(l)}$
for each layer l **do**
    Apply convolution operation: $Z^{(l)} = \sigma(W^{(l)} * X + b^{(l)})$
    Apply average pooling: $P^{(l)} = \text{pool}(Z^{(l)})$
end for
Initialize gated unit
for each time step t **do**
    Compute update gate: $z_t = \sigma(W_z X_t + U_z h_{t-1} + b_z)$
    Compute reset gate: $r_t = \sigma(W_r X_t + U_r h_{t-1} + b_r)$
    Compute candidate activation: $\tilde{h}_t = \tanh(W_h X_t + U_h(r_t \odot h_{t-1}) + b_h)$
    Compute hidden state: $h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$
end for
Concatenate convolutional and gated unit outputs: $M = \text{concat}(h_T, P^{(L)})$
Compute output probability: $\hat{y} = \sigma(W_o M + b_o)$

ConvRecurrentNet optimization using grasshopper optimization algorithm

We optimized the proposed CRNet by applying the GOA. It is a metaheuristic optimization technique inspired by the swarming behavior of grasshoppers in nature [26]. The movement of grasshoppers is mathematically modeled to explore and exploit the search space effectively, leading to optimal solutions for complex problems [27]. The position update equation for each agent in GOA is shown in Fig. 2 (2) given by:

$$\mathbf{X}_i^{t+1} = \mathbf{X}_i^t + \mathbf{S}_i + c(\mathbf{X}_g - \mathbf{X}_i^t) \tag{15}$$

where $\mathbf{X}_i^t$ represents the position of the ith grasshopper at iteration t, $\mathbf{S}_i$ is the social interaction component, $\mathbf{X}_g$ is the global best solution (leader), and c is a decreasing control parameter that balances exploration and exploitation. The social interaction term $\mathbf{S}_i$ is calculated as:

$$\mathbf{S}_i = \sum_{j=1, j\neq i}^{N} s(d_{ij})\hat{\mathbf{d}}_{ij} \tag{16}$$

where $s(d_{ij})$ is the strength of attraction or repulsion between the ith and j^{th} grasshoppers based on their Euclidean distance d_{ij}, and $\hat{\mathbf{d}}_{ij}$ is the unit vector in the direction of interaction. The function $s(d)$ is shown in Fig. 2 (2) defined as:

$$s(d) = fe^{-d/\ell} - e^{-d} \tag{17}$$

where f is the scaling factor and ℓ is the length scale parameter that controls the range of interaction between agents. The control parameter c decreases over iterations to transition from global search (exploration) in the early stages to local search (exploitation) in the later stages, ensuring convergence to an optimal solution. The value of c at each iteration as shown in Fig. 2 (3) is updated by:

$$c^t = c_{\max} - t\left(\frac{c_{\max} - c_{\min}}{T}\right) \tag{18}$$

where $c_{\max}$ and $c_{\min}$ are the initial and final values of c, the current iteration is t, and T is the maximum number of iterations. The optimization process iteratively updates the hyperparameters of the proposed CRNet model by evaluating candidate solutions and selecting the best-performing configuration based on the objective function, which is the classification accuracy in the current situation as shown in Fig. 2 (4).

The optimized hyperparameters of the CRNet model after using GOA are shown in Table 6. GOA draws inspiration from the swarming behavior of grasshoppers in the wild to fine-tune the hyperparameters for improved classification results. The range of values in Table 6 shows the potential intervals for every hyperparameter taken into account when the GOA was used for optimization [28]. These ranges delineate the search space where in GOA investigates the optimal configuration for enhancing the performance of CRNet. GOA makes sure that the chosen hyperparameters optimize classification accuracy while reducing overfitting by methodically modifying values like gated units (which range from 8 to 128), dense units (16 to 256), learning rate (0.0001 to 0.01), and dropout rate (0.2 to 0.5). Compared to the original configurations, the optimization process significantly modified the architecture by raising the number of dense

Table 6 Optimized hyperparameters

Hyperparameter	Values Range	Optimal Value
Gated Units	8–128	83
Dense Units	16–256	179
Filters	16–128	114
Kernel Size	3–7	6
Activation Function	ReLU, Tanh, Sigmoid	Sigmoid, ReLU
Optimizer	Adam, SGD, RMSprop	Adam
Learning Rate	0.0001–0.01	0.00274
Dropout Rate	0.2–0.5	0.2951

units to 179 and gated units to 83. By optimizing the dropout rate to 0.2951, overfitting is avoided, and regularization is enhanced. A kernel size of 6 and 114 filters are applied in the convolutional layers, improving feature extraction. The *Sigmoid* and *ReLU* activation functions are utilized, which offer the data non-linear transformations. The *Adam* optimizer is used to balance stability and convergence speed with a learning rate and dropout rate of 0.00274 and 0.2951. The CRNet model performed better due to these optimized hyperparameters, reducing prediction uncertainty and increasing classification accuracy.

GOA can avoid local optima, adapt to complex search spaces, and strike a strong balance between exploration and exploitation. Because of these advantages, GOA works incredibly well for enhancing DL models such as CRNet. The improved feature extraction and classification accuracy of the optimized CRNet model for predicting heart disease leads to more trustworthy diagnoses. GOA reduces prediction uncertainty by optimizing hyperparameters, which improves the model's capacity to identify minute patterns in medical data. In the end, this optimization helps detect heart disease earlier and more accurately, which supports clinical judgment and enhances patient outcomes in the medical field.

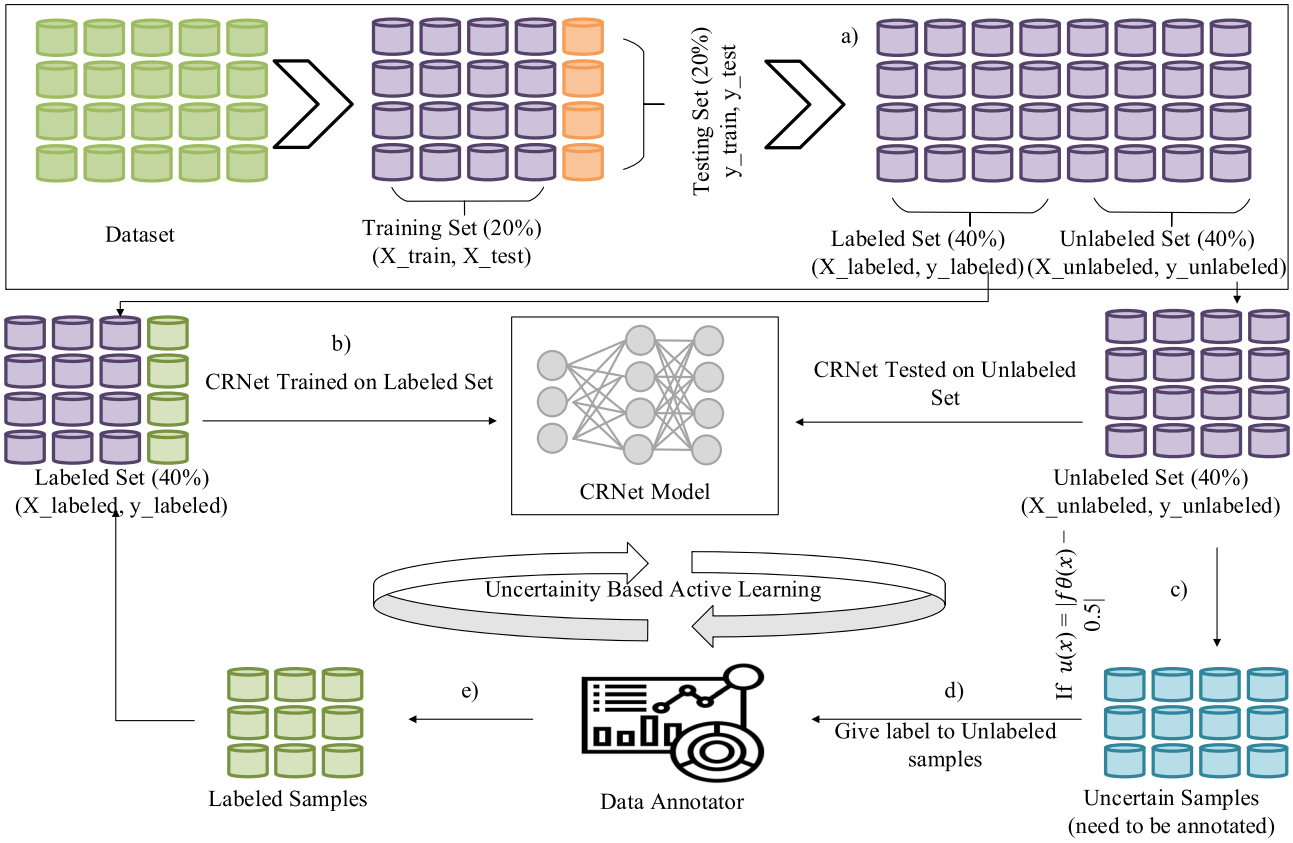


Fig. 4 Uncertainty based active learning flow diagram

Integration of ConvRecurrentNet with uncertainty-based active learning

Active learning is an effective technique to reduce labeling costs by selecting the most informative samples for annotation. Among various active learning strategies, UBS is widely used due to its simplicity and efficiency [29]. It focuses on selecting samples where the model exhibits the highest uncertainty in its predictions. This strategy enhances the learning process by improving the model's confidence in difficult-to-classify instances [30]. In an active learning framework, the dataset consists of a set of feature vectors X and their corresponding labels Y:

$$X = \{x_1, x_2, \ldots, x_n\}, \quad Y = \{y_1, y_2, \ldots, y_n\} \quad (19)$$

where X denotes the feature vectors and Y represents the corresponding binary labels $y_i \in \{0, 1\}$. The dataset is divided into labeled and unlabeled subsets as shown in Fig. 4 (a):

$$(X_{\text{labeled}}, Y_{\text{labeled}}), \quad (X_{\text{unlabeled}}, Y_{\text{unlabeled}}) \quad (20)$$

The CRNet model f_θ is trained on the labeled dataset as shown in Fig. 4 (b) to produce probabilistic predictions:

$$f_\theta(x) \in [0, 1] \quad (21)$$

The uncertainty of a sample is calculated by measuring its closeness to the decision boundary at 0.5 as shown in Fig. 4 (c), which is defined as:

$$u(x) = |f_\theta(x) - 0.5| \quad (22)$$

where lower values of $u(x)$ indicate higher uncertainty. The most uncertain samples are selected by sorting the unlabeled data according to their uncertainty scores as shown in Fig. 4 (d):

$$\mathcal{S}_{\text{uncertain}} = \text{argmin}_{x \in X_{\text{unlabeled}}}(u(x)) \quad (23)$$

where $\mathcal{S}_{\text{uncertain}}$ represents the most uncertain samples selected for labeling. $X_{\text{unlabeled}}$ is the set of unlabeled data points, and $u(x)$ is an uncertainty function that quantifies how uncertain the model is about the sample x. After that, the top k uncertain samples are chosen for labeling as shown in Fig. 4 (e):

$$\mathcal{S}_{\text{selected}} = \{x_1, x_2, \ldots, x_k\} \quad (24)$$

where $\mathcal{S}_{\text{selected}}$ represents the set of k samples chosen for labeling and $x_1, x_2, \ldots, x_k$ are the individual data points selected from the unlabeled dataset. These samples are transferred to the labeled dataset as shown in Fig. 4:

Table 7 Performance measures and formulas for accurate heart disease prediction at early stages

Measures	Formula
1. Accuracy	$\frac{TP+TN}{TP+TN+FP+FN} \times 100$
2. Precision	$\frac{TP}{TP+FP} \times 100$
3. Recall	$\frac{TP}{TP+FN} \times 100$
4. F1-Score	$2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \times 100$
5. ROC-AUC	$\int_{-\infty}^{\infty} \text{TPR}(t)\, d\text{FPR}(t)$
6. Cohen's Kappa	$\frac{P_o - P_e}{1 - P_e}$
7. MCC	$\frac{TP \times TN - FP \times FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$

$$X_{\text{labeled}}^{t+1} = X_{\text{labeled}}^{t} \cup \mathcal{S}_{\text{selected}}, \quad Y_{\text{labeled}}^{t+1} = Y_{\text{labeled}}^{t} \cup Y_{\text{selected}} \quad (25)$$

where X_{labeled}^{t+1} and Y_{labeled}^{t+1} represent the updated sets of labeled data points and their corresponding labels at iteration $t + 1$. X_{labeled}^{t} and Y_{labeled}^{t} are the labeled data points and their labels at iteration t. $\mathcal{S}_{\text{selected}}$ is the set of newly selected samples for labeling, and Y_{selected} contains their corresponding labels. Simultaneously, the selected samples are removed from the unlabeled dataset:

$$X_{\text{unlabeled}}^{t+1} = X_{\text{unlabeled}}^{t} \setminus \mathcal{S}_{\text{selected}} \quad (26)$$

The active learning process proceeds until a termination condition is met, such as reaching the maximum number of iterations, exhausting the labeling budget, or when only minimal performance improvement between consecutive iterations. UBS effectively improves model performance while minimizing the number of labeled instances, making it highly suitable for applications where labeled data is scarce and costly.

Simulations and results

In this section, we discussed the various performance measures and evaluated the simulation results of the proposed models. Table 7 offers the mathematical formulas for assessing heart disease prediction models performance. A percentage representing accuracy is obtained by adding the number of correctly predicted instances, True Positives (TP) and True Negatives (TN), by the total number of instances and then multiplying the result by 100 as given in Table 7 (Equation 1). Indicating the model's capacity to prevent False Positives (FP), precision calculates the percentage of TP predictions among all positive predictions as mentioned in Table 7 (Equation 2). Conversely, recall highlights the model's capacity to identify positive cases by calculating the percentage of TP predictions among all actual positive instances as given in Table 7 (Equation 3). The F1-Score, which is the harmonic mean of accuracy and recall, finds

Table 8 Comparison of confusion matrices for CRNet (with and without OUPS)

Actual Class	Without Balancing			With Balancing		
	Absent	**Present**	**Total**	**Absent**	**Present**	**Total**
Absent TN/FP	**22830 (90.0%)**	153 (0.60%)	22983	**22225 (48.36%)**	686 (1.49%)	22911
Present FN/TP	2185 (8.61%)	**200 (0.79%)**	2385	3300 (7.18%)	**19748 (42.97%)**	23048
Total Predicted	25015 (98.6%)	353 (1.4%)	25368	25525 (55.5%)	20434 (44.5%)	45959

Table 9 Comparison of confusion matrices for optimized CRNet (with and without OUPS)

Actual Class	Without Balancing			With Balancing		
	Absent	**Present**	**Total**	**Absent**	**Present**	**Total**
Absent TN/FP	**22850 (90.07%)**	133 (0.52%)	22983	**22414 (48.77%)**	369 (0.80%)	22783
Present FN/TP	2195 (8.65%)	**190 (0.75%)**	2385	3511 (7.64%)	**19665 (42.79%)**	23176
Total Predicted	25045 (98.7%)	323 (1.3%)	25368	25925 (56.0%)	20034 (44.0%)	45959

Table 10 Comparison of confusion matrices for CRNet with UBS (with and without OUPS)

Actual Class	Without Balancing			With Balancing		
	Absent	**Present**	**Total**	**Absent**	**Present**	**Total**
Absent TN/FP	**22575 (88.99%)**	351 (1.38%)	22926	**22499 (48.95%)**	404 (0.88%)	22903
Present FN/TP	2071 (8.16%)	**371 (1.46%)**	2442	2369 (5.15%)	**20687 (45.01%)**	23056
Predicted Total	24645 (97.15%)	722 (2.85%)	25367	24868 (54.11%)	21091 (45.89%)	45959

a balance between these two metrics, as seen in Table 7 (Equation 4). The model's performance relies heavily on these measures, especially when dealing with imbalanced datasets. The model's capacity to discriminate between classes is measured by the Receiver Operating Characteristic-Area Under the Curve (ROC-AUC), a higher value denotes superior discrimination as demonstrated in Table 7 (Equation 5). Considering the random correlation between the labels that are assigned and the predictions that are made, Cohen's Kappa offers a more accurate assessment of classification performance as given in Table 7 (Equation 6). Suitable for imbalanced datasets, the Matthews Correlation Coefficient (MCC) evaluates prediction quality by considering each of the four confusion matrix elements shown in Table 7 (Equation 7). Log loss penalizes incorrect classifications with higher confidence more severely, measuring the uncertainty of probabilistic predictions.

Experimental setup

All experiments are performed on Google Colab for model development and training, using the latest runtime environment available at the time (Python 3.10). The computational setup used for execution is a machine running Windows 10, powered by an Intel(R) Core(TM) i5-7300 U processor operating at 2.60 GHz (2.71 GHz).

Performance analysis of data balancing

The confusion matrices reported in Tables 8, 9, and 10 provide valuable insights into the classification ability of CRNet, optimized CRNet, and CRNet with UBS, both with and without balancing using OUPS. Beginning with Table 8, which corresponds to CRNet, we see that without balancing the model correctly identifies $TN = 22830$ healthy individuals and $TP = 200$ diseased individuals, while misclassifying $FP = 153$ healthy samples as diseased and $FN = 2185$ diseased samples as healthy. These results reveal a strong bias toward the majority class since the dataset is highly imbalanced, leading to a very large number of FN. After balancing, however, the situation improves considerably, where the model produces $TN = 22225$, $TP = 19748$, $FP = 686$, and $FN = 3300$. The increase in TPs from 200 to 19748 shows that balancing greatly enhances the sensitivity of the model toward minority class detection, even though this comes at the expense of a slight rise in FPs and FN due to increased exposure to diverse synthetic samples. Overall, OUPS corrects the bias and allows CRNet to recognize diseased cases more effectively.

For Table 9, which presents the results of optimized CRNet, a similar trend is observed but with notable improvements in classification quality. Without balancing, the model records $TN = 22850$, $TP = 190$, $FP = 133$, and $FN = 2195$. Although the number of FPs is slightly lower than in the original CRNet, the number of FNs remains excessively high, again reflecting bias toward the majority class. After balancing, the optimized model demonstrates $TN = 22414$, $TP = 19665$, $FP = 369$, and $FN = 3511$. The significant rise in TPs from only 190 to 19665 illustrates the effectiveness of balancing combined with hyperparameter optimization through GOA, which allows the model to better capture informative features of the minority class. The moderate increase in FNs is acceptable since the majority of diseased cases are now identified, and the reduced FPs compared to CRNet confirm improved robustness.

Finally, Table 10 shows the performance of CRNet with UBS. Without balancing, the model yields $TN = 22575$, $TP = 371$, $FP = 351$, and $FN = 2071$. Even in the unbalanced setting, the active learning strategy that prioritizes uncertain samples helps the model capture a considerable number of positive cases, refining its decision boundaries around challenging instances. When balancing is applied, the performance changes to $TN = 22499$, $TP = 20687$, $FP = 404$, and $FN = 2369$. Although the number of TNs slightly decreases because the dataset itself is balanced and thus contains fewer majority instances, the number of correctly identified diseased cases (TP) increases dramatically from 371 to 20687. This improvement demonstrates that the combination of balancing and active learning synergistically enhances sensitivity while maintaining strong specificity, ensuring the model generalizes effectively across both classes.

In summary, without balancing, all models suffer from high FNs due to majority class bias, except CRNet with UBS which already mitigates this problem by focusing on uncertain samples. With balancing, all models show dramatic improvements in minority class recognition, with CRNet with UBS achieving the best trade-off between TPs and TNs. The progression from CRNet to optimized CRNet and finally CRNet with UBS highlights how balancing, optimization, and active learning complement each other in improving classification performance on imbalanced datasets.

Overfitting analysis

In DL, overfitting is a prevalent problem where a model learns patterns that are too specific to the training set, resulting in poor generalization to unseen data. This usually appears as a significant discrepancy between training and validation loss, where the former keeps declining while the latter either plateaus or rises. Overfitting often arises from insufficient data, lack of regularization, or an overly complex model. Techniques such as dropout regularization, early stopping, and hyperparameter tuning are commonly employed to reduce overfitting. The study's active learning architecture and optimized hyperparameter tuning minimize overfitting, as seen in Fig. 5. It compares the training and validation losses of CRNet, Optimized CRNet, and CRNet with UBS.

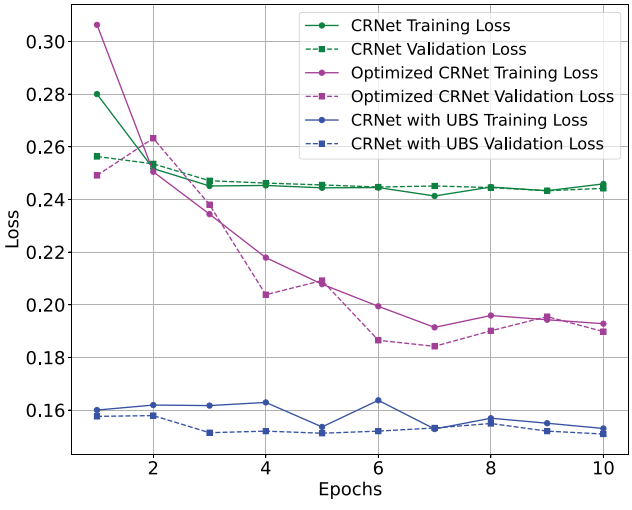


Fig. 5 Loss comparison of CRNet, optimized CRNet, and CRNet with UBS

The improved hyperparameter tuning of the optimized CRNet model includes more dense and gated units as well as a fine-tuned learning rate through the use of GOA. These enhancements enable the optimized CRNet to converge more efficiently than CRNet, reducing training and validation loss. Consistently lower validation loss across epochs confirms that the optimized model is not only learning more effectively but also generalizing better to unseen data. Figure 5's loss curves show that both models begin at epoch 1 with comparatively high loss values. However, the loss for both models decreases as training progresses. The faster loss decline in the first few epochs demonstrates the efficiency of the optimized CRNet's improved architecture and hyperparameter adjustments. By epoch 4, the difference between the two models is more noticeable, with optimized CRNet maintaining a much lower loss than CRNet. The persistence of this pattern indicates that the model architecture enhancements lead to a more effective learning process. The loss of the CRNet model starts to stabilize as training progresses to later epochs, especially around epoch 6, although it is still higher than the optimized CRNet. Small variations in the CRNet validation loss point to possible overfitting or instability in generalization.

The optimized CRNet, on the other hand, maintains a consistent and seamless drop in validation loss, demonstrating its enhanced generalization and robustness. While CRNet maintains a relatively higher loss level, the optimized CRNet reaches its lowest loss value by epoch 10, the last training epoch. Like the other models, the CRNet model with UBS starts with a loss value at epoch 1. However, because UBS focuses on the most informative samples, the model starts learning from the most informative samples at the outset. Compared to the standard and optimized CRNet versions, the model can extract meaningful patterns more quickly due to this targeted learning strategy. Even at this early stage of training, all models show high loss; however, CRNet with UBS exhibits a somewhat sharper decline in loss, suggesting a more stable learning process. By epoch 5, the CRNet with UBS has already achieved significantly decreased loss. Faster convergence results from the model's improved ability to refine its decision boundaries due to its emphasis on uncertain samples. CRNet with UBS is still improving at epoch 6. The model has absorbed the majority of

the key patterns from the dataset, as evidenced by the initial sharp drop in loss. The validation loss is consistently lower for the UBS-based model than for the CRNet and optimized CRNet, which may still be fine-tuning their decision boundaries. This stability shows that the model has successfully adjusted to the dataset and is no longer learning from redundant or less informative samples. The CRNet with UBS demonstrates its superior learning efficiency by achieving the lowest loss of all three models by epoch 10. The UBS-based model keeps a steadier trajectory than the conventional CRNet, which still shows variations in validation loss. This shows that the model is not only learning efficiently but also improving its ability to generalize to new data. The risk of overfitting lowers because the gap between training and validation loss constantly narrows.

Justification for benchmark models

The main reasons for choosing DL models as benchmark models are their unique features and capacity to provide a thorough performance evaluation. Each model possesses unique advantages and disadvantages, making them suitable benchmarks for evaluating the performance of the proposed CRNet framework. Furthermore, the existing models are chosen due to their similar architecture and functionality to my proposed model, CRNet. A more complete picture of CRNet's predictive power and general robustness can be acquired by contrasting it with these models.

- GRU is selected as a benchmark model because of its gated mechanism, which successfully resolves the vanishing gradient issue and enables it to capture long-range dependencies in sequential data [31]. It is ideal for tasks involving temporal patterns like medical time-series prediction because of its capacity to selectively remember and forget information. GRUs inability to extract spatial features and local dependencies from the data is due to its lack of convolutional layers. This limitation reduces its effectiveness when handling intricate patterns that call for hierarchical feature extraction in addition to sequential learning. The capabilities of convolutional layers and gated recurrent units are combined in CRNet, which enables it to recognize spatial correlations in the input data and learn sequential dependencies.
- LeNet is selected because of convolutional layers that can effectively detect local patterns and extract spatial features from input data. It is able to acquire significant representations through hierarchical feature extraction, which is essential for classification tasks [32]. Although originally designed for image data, its convolutional feature extraction capability is also valuable for structured tabular health indicators where local dependencies between features exist [33]. However, LeNet's inability to model sequential dependencies and retain long-term contextual information is due to its lack of a gated mechanism. LeNet is less suited for tasks requiring sequential learning since it is unable to adequately capture temporal patterns in data in the absence of a recurrent structure. By combining convolutional layers with a gated mechanism, CRNet gets around this restriction and is able to capture spatial patterns while maintaining temporal dependencies. Recurrent layers improve the model's capacity to remember important sequential information, guaranteeing that both local and long-range dependencies are efficiently learned.
- Echo State Network (ESN) reservoir layers are notable for their ability to map input signals into high-dimensional representations in addition to their computational efficiency [34]. This feature improves the model's capacity to distinguish between sequential data's linearly inseparable patterns. Although ESN is typically employed for time-series tasks, it is included here as a benchmark since its recurrent reservoir dynamics allow exploration of how sequential modeling techniques adapt when applied to structured tabular health data. However, ESN fixed random reservoir weights pose a serious drawback, limiting its applicability to complex and varied datasets. The lack of updating of these weights during training makes it difficult for the model to learn from various data distributions in an optimal manner, which limits its ability to handle intricate medical prediction tasks. On the other hand, CRNet uses trainable recurrent layers, which enable it to modify and improve its learned representations in real time according to the dataset. Its capacity to generalize across various situations is improved by this flexibility, which raises the classification of heart disease prediction accuracy and robustness.
- DNN is a potent benchmark as it can learn intricate non-linear mappings from data due to its universal approximation capability [35]. DNNs can capture complex feature interactions with great flexibility. However, because DNNs do not have a specific mechanism for capturing temporal dependencies, they are intrinsically less effective for sequential data. Standard DNNs handle inputs independently, which makes them less appropriate for tasks requiring sequential learning than recurrent models, which are specifically made to retain prior knowledge. This drawback lessens their usefulness in applications like heart disease prediction from time-series medical data where temporal relationships are important.

Table 11 Insights on Model performance

Model (Accuracy (%))	Insights
LeNet (86)	Good at capturing spatial features but lacks sequential dependency handling.
GRU (87)	Efficient in capturing temporal dependencies but struggles with feature extraction.
LSTM (69)	Capable of modeling long-term dependencies, but suffers from high computational cost and overfitting on limited data.
ESN (86)	Uses reservoir computing for sequential learning but lacks trainable recurrent layers, limiting adaptability.
DNN (80)	Performs well in non-sequential tasks but lacks specialized mechanisms for spatial and temporal feature extraction.
CRNet (90)	Combines spatial and sequential patterns, providing superior classification accuracy.
Optimized CRNet (92)	Improves CRNet performance through GOA by fine-tuning hyperparameters.
CRNet with UBS (94)	Combines CRNet with UBS to select the most uncertain samples, enhancing performance with minimal labeled data.

By combining both convolutional and recurrent layers, CRNet overcomes this limitation and is able to identify temporal and spatial dependencies in the data. Important spatial features are extracted by the convolutional layers, and a more thorough feature representation is ensured by the recurrent components, which aid in maintaining sequential patterns.

Validation results of ConvRecurrentNet deep model

This section explains the proposed model simulation results. We examine the efficacy of the proposed models in predicting heart disease and provide a thorough performance comparison with existing models. We conducted the simulations several times for every model and recorded the best results for performance assessment. The insights of existing models and proposed models are given in Table 11. Different performance metrics are used to compare the performance of several models, including GRU, LSTM, LeNet, ESN, DNN, CRNet, Optimized CRNet, and CRNet with UBS in Table 12 and Fig. 6. According to the findings, CRNet with UBS performs better than any other model on 3 iterations. The model's overall effectiveness in predicting heart disease is demonstrated by its highest accuracy of 94%. With 98% precision it shows that its forecasts are extremely trustworthy. The 90% recall score demonstrates the models can detect cases of heart disease. Furthermore, CRNet with UBS achieves the highest F1-score of 99%, demonstrating its balanced performance concerning to recall and precision. These outcomes demonstrate that the CRNet model's learning process is improved by the incorporation of UBS, increasing its efficacy and efficiency in the prediction of heart disease.

The optimized CRNet model has the highest accuracy at 92%. With a 90% accuracy rate, CRNet comes next, followed by GRU with 87%, LeNet and ESN with 86%, and DNN with 80%. Meanwhile, LSTM lags behind with only 69% accuracy, highlighting its limited ability to capture temporal dependencies effectively in this dataset. This illustrates the benefit of the CRNet architecture, particularly when it is optimized using the GOA, which adjusts the model's hyperparameters for better performance. When it comes to precision, the optimized CRNet model scores 93%, while the CRNet model achieves 90%. While ESN attains 87%, DNN has the lowest precision at 80%, while GRU and LeNet both reach 87%. In comparison, LSTM records a much lower precision of 73%, which implies that it generates relatively more FPs compared to other models. This suggests that when compared to other models, optimized CRNet produces fewer FP predictions. The optimized CRNet model scores 91% for recall, which gauges the model's capacity to accurately identify cases of heart disease. With an 80% recall, DNN once more exhibits the worst performance. LSTM performs even worse with only 59% recall, indicating that it fails to detect a significant portion of TP cases, making it less reliable for medical prediction tasks. Recall improvements for CRNet and optimized CRNet indicate that these models are more effective at identifying real cases of heart disease and lowering FN. At 92% optimized, CRNet has the highest F1-score, which strikes a balance between precision and recall, followed by CRNet at 90%. DNN maintains its 80% score. GRU and ESN both achieve 86%, while LeNet achieves 84%. LSTM, however, attains only 65% F1-score, reflecting its poor balance between precision and recall. This reinforces the conclusion that LSTM underperforms significantly compared to CRNet variants and even baseline models. The better balance between detecting positive cases and preventing FP is confirmed by the higher F1-score of CRNet and optimized CRNet.

The log loss comparison of several models, including GRU, LSTM, LeNet, ESN, DNN, CRNet, and optimized CRNet, are shown in Fig. 7. Log loss, referred to as cross-entropy loss or logistic loss, measures the degree of uncertainty in a model's predictions. Higher log loss values imply that the model has trouble with classification uncertainty, whereas lower values show more certain and accurate predictions. According to Fig. 7, the LSTM model exhibits the worst performance, with the highest log loss of 0.687. This markedly high loss suggests that LSTMs predicted probabilities are poorly calibrated for this imbalanced, largely tabular problem—likely due to a mismatch between its sequence-oriented architecture and non-sequential feature structure, leading to

Table 12 Performance comparison of CRNet, optimized CRNet, CRNet with UBS, and existing models (10-fold cross validation)

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	ROC-AUC (%)	MCC (%)	Cohen's Kappa (%)	Log Loss	Training Time (s)
GRU	**87**, 86, 88, 87, 85, 87, 86, 87, 87, 86, 88, **87**	**87**, 85, 86, 87, 86, 88, 87, 86, 88, 87, 86, **87**	**84**, 83, 85, 84, 83, 84, 84, 85, 84, 83, 84, **84**	**86**, 85, 87, 86, 85, 86, 87, 86, 86, 86, 85, **86**	**95**, 94, 95, 94, 94, 95, 95, 94, 95, 95, 94, **95**	**74**, 73, 74, 75, 73, 74, 74, 73, 74, 75, 74, **74**	**74**, 72, 75, 74, 73, 74, 73, 74, 74, 73, 75, **74**	**0.275**, 0.278, 0.276, 0.280, 0.277, 0.275, 0.276, 0.278, 0.276, 0.274, 0.273, **0.275**	**383**, 375, 390, 385, 380, 384, 378, 376, 382, 387, 388, **383**
LSTM	**69**, 68, 68, 67, 66, 67, 65, 67, 67, 68, 64, **67**	**73**, 72, 70, 70, 72, 74, 72, 73, 76, 74, 72, **73**	**59**, 53, 54, 54, 53, 54, 54, 57, 52, 51, 57, **57**	**65**, 65, 66, 65, 65, 67, 66, 67, 64, 63, 61, **63**	**72**, 74, 71, 75, 72, 75, 75, 74, 70, 76, 70, **71**	**38**, 36, 31, 38, 37, 34, 36, 33, 34, 35, 32, **37**	**39**, 32, 35, 36, 38, 39, 30, 31, 38, 31, 36, **38**	**0.687**, 0.543, 0.642, 0.659, 0.682, 0.681, 0.683, 0.684, 0.687, 0.689, 0.686, **0.685**	**384**, 350, 364, 369, 360, 374, 379, 382, 382, 385, 384, **384**
LeNet	**86**, 85, 87, 86, 85, 86, 87, 86, 86, 85, 87, **86**	**87**, 85, 86, 86, 85, 87, 86, 86, 87, 85, 86, **87**	**86**, 84, 85, 85, 84, 86, 85, 85, 85, 84, 86, **86**	**84**, 82, 85, 84, 83, 84, 85, 84, 84, 83, 85, **84**	**96**, 95, 95, 95, 96, 95, 96, 95, 96, 96, 95, **96**	**75**, 74, 75, 74, 73, 75, 75, 74, 75, 74, 74, **75**	**73**, 72, 73, 73, 72, 73, 73, 72, 73, 73, 73, **73**	**0.281**, 0.283, 0.282, 0.280, 0.281, 0.282, 0.283, 0.281, 0.282, 0.280, 0.281, **0.281**	**158**, 155, 160, 157, 159, 161, 156, 158, 157, 160, 159, **158**
ESN	**86**, 85, 87, 86, 86, 86, 87, 85, 86, 85, 87, **86**	**87**, 84, 86, 86, 85, 87, 86, 87, 87, 88, 87, **87**	**86**, 85, 86, 84, 86, 86, 85, 85, 87, 85, 86, **86**	**86**, 85, 86, 84, 85, 84, 86, 84, 84, 86, 84, **84**	**86**, 83, 85, 85, 86, 85, 86, 85, 86, 86, 85, **86**	**73**, 73, 74, 73, 75, 74, 73, 73, 75, 74, 73, **75**	**73**, 72, 73, 73, 72, 73, 73, 72, 73, 73, 73, **73**	**0.294**, 0.293, 0.286, 0.282, 0.284, 0.280, 0.283, 0.291, 0.292, 0.292, 0.293, **0.281**	**923**, 925, 910, 937, 929, 941, 923, 924, 917, 913, 923, **925**
DNN	**80**, 79, 80, 81, 80, 80, 80, 81, 79, 80, 80, **80**	**80**, 79, 81, 79, 80, 80, 80, 81, 79, 80, 80, **80**	**80**, 79, 81, 75, 80, 80, 80, 81, 79, 80, 80, **80**	**80**, 79, 81, 81, 80, 80, 80, 81, 79, 80, 80, **80**	**80**, 79, 81, 78, 80, 80, 80, 81, 79, 80, 80, **80**	**60**, 59, 61, 63, 60, 60, 60, 61, 59, 60, 60, **60**	**61**, 60, 62, 60, 61, 61, 61, 62, 60, 61, 61, **61**	**0.416**, 0.418, 0.412, 0.415, 0.419, 0.417, 0.416, 0.418, 0.417, 0.416, 0.415, **0.416**	**77**, 75, 73 78, 76, 77, 78, 76, 77, 76, 78, **77**
CRNet Model	**90**, 89, 87, 91, 90, 90, 91, 89, 90, 90, 89, **90**	**90**, 89, 91, 90, 91, 90, 91, 89, 90, 90, 89, **90**	**90**, 89, 91, 90, 89, 90, 91, 89, 90, 90, 89, **90**	**90**, 89, 91, 90, 90, 90, 91, 89, 90, 90, 89, **90**	**97**, 96, 97, 96, 95, 97, 97, 96, 97, 96, 97, **97**	**81**, 80, 82, 81, 80, 81, 82, 80, 81, 81, 80, **81**	**83**, 82, 84, 83, 82, 83, 84, 82, 83, 83, 82, **83**	**0.210**, 0.212, 0.211, 0.209, 0.213, 0.211, 0.210, 0.212, 0.211, 0.210, 0.209, **0.210**	**392**, 390, 391, 395, 393, 391, 394, 390, 392, 391, 393, **392**
Optimized CRNet	**92**, 91, 92, 90, 93, 91, 92, 92, 92, 93, 91, **92**	**93**, 92, 94, 91, 93, 92, 93, 93, 94, 92, 93, **93**	**91**, 90, 91, 89, 92, 90, 91, 91, 91, 92, 90, **91**	**92**, 91, 92, 87, 92, 91, 92, 92, 92, 92, 91, **92**	**98**, 97, 98, 90, 97, 98, 98, 97, 98, 98, 97, **98**	**82**, 81, 82, 85, 83, 81, 82, 82, 82, 83, 81, **82**	**84**, 83, 85, 83, 84, 83, 84, 84, 84, 85, 83, **84**	**0.194**, 0.195, 0.192, 0.193, 0.196, 0.194, 0.195, 0.193, 0.194, 0.194, 0.195, **0.194**	**763**, 760, 762, 765, 762, 761, 764, 761, 763, 762, 765, **763**
CRNet with UBS	**94**, 93, 90, 95, 94, 94, 93, 94, 94, 95, 93, **94**	**98**, 97, 93, 98, 97, 98, 98, 97, 98, 98, 97, **98**	**90**, 89, 90, 91, 90, 90, 89, 90, 90, 91, 89, **90**	**99**, 98, 94, 99, 98, 99, 99, 98, 99, 99, 98, **99**	**99**, 98, 97, 99, 98, 99, 99, 98, 99, 99, 98, **99**	**88**, 87, 85, 89, 88, 88, 87, 88, 88, 89, 87, **88**	**88**, 87, 86, 89, 88, 88, 87, 88, 88, 89, 87, **88**	**0.149**, 0.150, 0.149, 0.148, 0.151, 0.149, 0.150, 0.148, 0.149, 0.149, 0.150, **0.149**	**385**, 382, 384, 385, 385, 384, 385, 385, 383, 385, 384, **385**

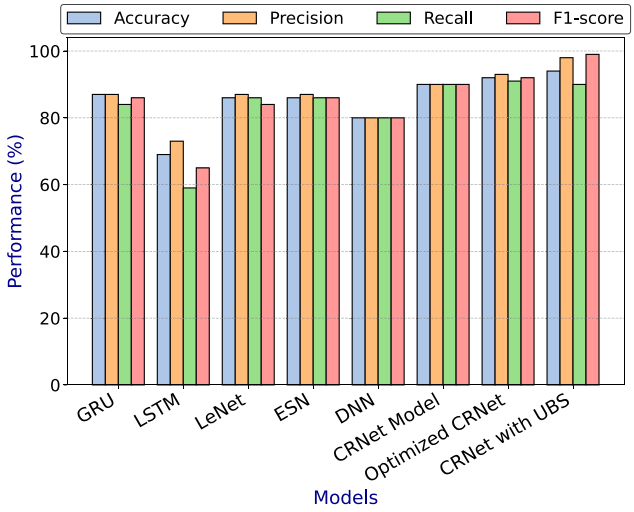


Fig. 6 Performance comparison of CRNet, optimized CRNet, CRNet with UBS, and existing models

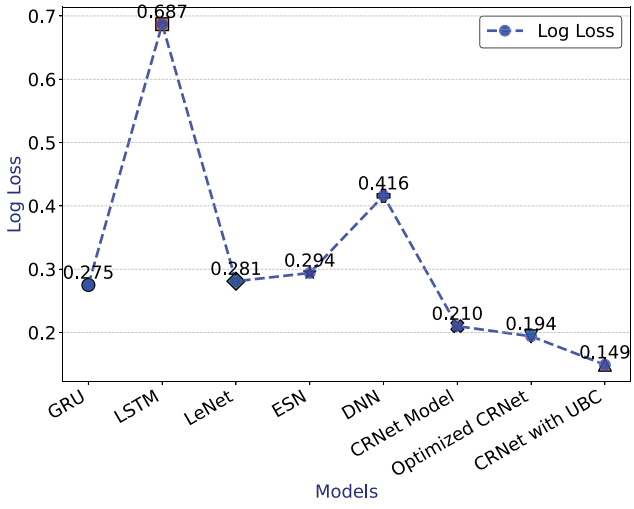


Fig. 7 Log loss comparison of CRNet, optimized CRNet, CRNet with UBS, and existing models

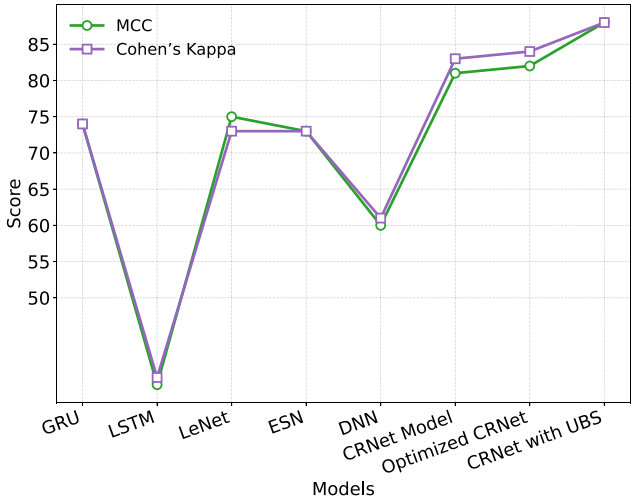


Fig. 8 Cohen kappa and MCC comparison of CRNet, optimized CRNet, CRNet with UBS, and existing models

underfitting of discriminative patterns; sensitivity to the extreme class imbalance, which yields overconfident mistakes that log loss penalizes heavily; and the absence of convolutional feature extractors that, as in CRNet, help learn robust local patterns and produce better-calibrated probabilities. The DNN model also performs poorly at differentiating between positive and negative cases, as evidenced by a high log loss of 0.416. This high log loss indicates that in comparison to other models, DNN produces predictions with lower confidence and misclassifies more cases. The main cause of this subpar performance is that DNN is less capable of capturing intricate relationships within the dataset because it lacks specialized feature extraction mechanisms like convolutional layers or recurrent units. ESN achieves a log loss of 0.294, which is still a little higher than DNNs. Although reservoir computing is incorporated into ESN feature learning and classification, they are not optimally balanced, which increases prediction uncertainty. In a similar vein, the log loss values for the LeNet and GRU models are 0.281 and 0.275, respectively. Compared to ESN and DNN, these models perform better; however, their higher loss values suggest that they have trouble maintaining accurate classification confidence levels. As a recurrent model, GRU works well for sequential data but cannot be the best at classifying tabular medical data. LeNet has a slightly higher log loss because it was not fully optimized for structured datasets when it was first developed for image processing. CRNet produces much more confident and dependable predictions, as evidenced by its much lower log loss of 0.210. Convolutional layers aid in the extraction of significant features, and gated recurrent components guarantee that the data's sequential dependencies are successfully recorded. With the help of this architecture, CRNet can produce more precise probability estimates and generalize to new instances more effectively. Optimized CRNet attains the lowest log loss at 0.194. The GOA, which adjusts hyperparameters like learning rate, dropout rate, the number of filters, and units in the model, is responsible for the improvement over CRNet. To reduce prediction uncertainty and produce more certain classification decisions, GOA improves the model's capacity to learn optimal feature representations. With the lowest log loss value of 0.149, the CRNet with UBS produces predictions with a high degree of confidence and significantly reduced uncertainty. The model's ability to assign higher probabilities to correct classes is reflected in a lower log loss value, which increases the predictability of the model. By integrating UBS, CRNet concentrates on the most ambiguous and instructive

samples, improving learning and lowering uncertainty in decision-making.

The comparison of Cohen's Kappa and MCC for various models, such as GRU, LSTM, LeNet, ESN, DNN, CRNet, and Optimized CRNet, is shown in Fig. 8. In imbalanced classification problems, such as the prediction of heart disease, these metrics offer a more thorough understanding of model performance than conventional accuracy measures. When assessing the degree of agreement between expected and actual labels, Cohen's Kappa considers random chance. Improved model reliability is indicated by higher values. Among all models, LSTM performs the worst with a Cohen's Kappa score of only 39% and an MCC of 38%. These very low values indicate that LSTM fails to achieve meaningful agreement with the ground truth and struggles to maintain balanced performance across classes. The poor results can be attributed to LSTMs architectural bias towards sequential dependencies, which does not align well with the structured, tabular medical dataset used here. Additionally, its inability to effectively handle extreme class imbalance leads to higher FNs, significantly reducing both MCC and Cohen's Kappa. At 61%, the DNN model has the lowest Cohen's Kappa score among the baseline models (excluding LSTM), suggesting that its predictions and the labels it uses don't agree well. Given the comparatively high log loss and lower accuracy of DNN, this low score is to be expected. Even though ESN and LeNet perform marginally better achieving MCC of 73 and 75%, and Cohen's Kappa scores of 73%, respectively, they are still not very good at making predictions, most likely because they are not very good at identifying significant features in structured medical data. At a Cohen's Kappa and MCC score of 74%, the sequential model GRU outperforms these baseline models. In contrast to more specialized architectures, its performance plateaus because its recurrent nature might not be entirely effective for tabular datasets. CRNet, which combines recurrent and convolutional mechanisms, performs noticeably better than these models with a Cohen's Kappa and MCC scores of 83 and 81%. The enhanced generalization and robustness of CRNet are demonstrated by the greater degree of agreement between its predictions and the actual labels. With a Cohen's Kappa and MCC scores of 84 and 82%, the optimized CRNet exhibits even greater consistency and dependability. Its parameters have been improved with the use of the GOA, producing a model that minimizes misclassification errors and produces predictions that are more accurate and certain. CRNet with UBS has the highest Cohen's Kappa score of 88%, demonstrating the model's high prediction consistency. This suggests that even when there is imbalanced data, the model produces extremely accurate predictions. Furthermore, CRNet with UBS achieves the highest MCC score of 88%, demonstrating its performance balance across all classes. The MCC value highlights the model's ability to maintain consistent performance without being biased toward majority or minority classes. The better performance of CRNet with UBS in both metrics shows that incorporating UBS into the model improves its overall robustness and reliability in predicting heart disease in addition to improving classification accuracy.

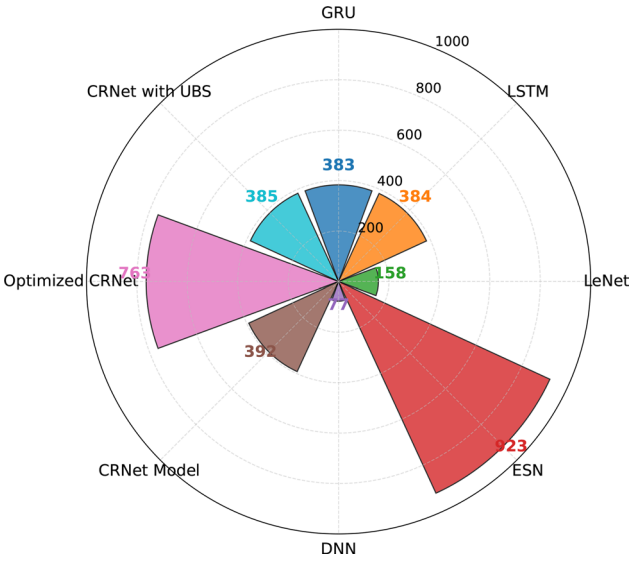


Fig. 9 Execution time comparison of CRNet, optimized CRNet, CRNet with UBS, and existing models

The execution time comparison of existing and proposed models is shown in Fig. 9. In real-time medical applications where quicker predictions can help with prompt diagnosis and treatment, execution time is a crucial consideration when assessing a model's computational efficiency. DNN is the quickest model of all, with the smallest execution time of 77 seconds. This is to be expected given that DNN is computationally light due to its lack of intricate feature extraction mechanisms. LeNet with a 158 seconds execution time, which is marginally longer because of its convolutional layers, which increase processing overhead. GRU takes 383 seconds, showing that because sequential dependencies necessitate more intricate computations, its recurrent structure lengthens computation time. LSTM requires 384 seconds of execution time, which is nearly the same as GRU. This longer training time compared to lightweight models like DNN or LeNet arises from LSTMs gating mechanisms (input, forget, and output gates), which introduce additional computational complexity at each timestep. However, because the dataset is primarily tabular and not sequential, these extra computations do not translate into meaningful performance gains, making LSTM both slower and less effective. The ESN has a much longer execution time of 923 seconds. The reservoir computing aspect of ESN

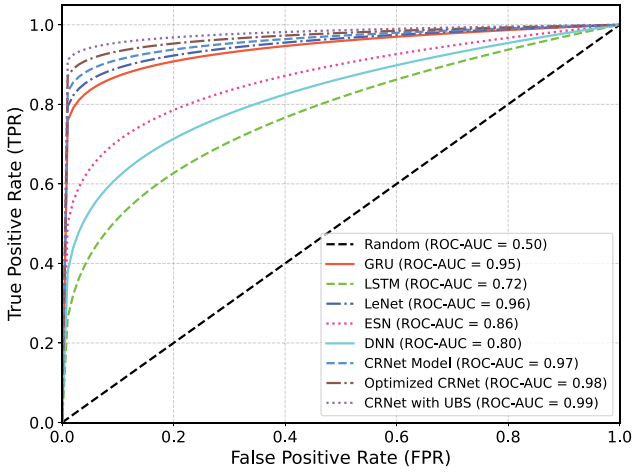


Fig. 10 ROC-AUC comparison of CRNet, optimized CRNet, CRNet with UBS, and existing models

adds to the processing time by requiring more training time. An execution time of 392 seconds for CRNet, which combines convolutional and recurrent components, is higher than GRU but lower than ESN, suggesting a balance between increased performance and computational cost. The execution time, 763 seconds, is achieved by optimized CRNet, which uses the GOA for hyperparameter tuning. The extra optimization steps needed to fine-tune parameters that improve accuracy and reduce loss are the cause of this increase in computation time. There is a trade-off between accuracy and efficiency because an optimized CRNet produces better results despite taking longer to train. CRNet with UBS attained an execution time of 385 seconds, which is the lowest among the proposed models. This demonstrates that the incorporation of UBS not only enhances predictive performance but also significantly reduces computational cost. By selecting the most uncertain and informative samples, UBS decreases the number of training instances required, thereby accelerating the learning process while maintaining high accuracy.

The ROC-AUC curves for existing and proposed models are shown in Fig. 10. The trade-off between the FP rate (1-specificity) and the TP rate (sensitivity) at various classification thresholds is represented by the ROC-AUC curve. Better discriminatory power is indicated by higher values of the ROC-AUC, a crucial metric that measures the model's capacity to differentiate between positive and negative classes. Random guessing is shown by the baseline reference line at ROC-AUC of 0.50. LSM performs the worst of all with an ROC-AUC of 0.72, indicating limited predictive capability and only slightly better than random classification. DNN performs the worst of the models with an ROC-AUC of 0.80, indicating that it has a limited capacity for prediction and is only marginally superior to a random classifier. ESN performs moderately, trailing more sophisticated architectures with an ROC-AUC of 0.86. Though it is primarily intended for image-based tasks rather than tabular data, LeNet performs marginally better, reaching 0.96, demonstrating its effectiveness in identifying cases of heart disease. Because it can identify sequential dependencies, GRU exhibits strong predictive power with an ROC-AUC of 0.95. However, it is outperformed by CRNet, which has an ROC-AUC of 0.97. CRNet can extract more significant features from structured data by incorporating convolutional layers, which increases classification accuracy and lowers FP. With an ROC-AUC of 0.98, the optimized CRNet is the model that performs the best. When the GOA is used, its feature selection and classification accuracy are improved, resulting in almost flawless differentiation between healthy and diseased people. The small gap between CRNet and optimized CRNets indicates that changing hyperparameters can improve model performance. The greatest ROC-AUC value of 0.99 is attained by CRNet with UBS, demonstrating its remarkable

Table 13 Architecture utilized in CRNet

Architecture
1. Input Layer (shape = $X_{train} \times 1$)
2. Gated Layer (Units = 16, Activation = 'tanh')
3. Dense Layer (Units = 32, Activation = 'tanh')
4. Dense Layer (Units = 16, Activation = 'tanh')
5. Input Layer (shape = $X_{train} \times 1$)
6. Convolution Layer (Filters = 64, Kernel size = 5, Activation = 'Sigmoid')
7. Average Pooling Layer (Pool size = 2)
8. Convolution Layer (Filters = 64, Kernel size = 5, Activation = 'tanh')
9. Average Pooling Layer (Pool size = 2)
10. Flatten Layer
11. Dense Layer (Units = 120, Activation = 'tanh')
12. Dense Layer (Units = 84, Activation = 'tanh')
13. Concatenation Layer
14. Dense Layer (Units = 1, Activation = 'Sigmoid')

discriminatory power. This outcome demonstrates the model's high accuracy in consistently differentiating between positive and negative classes. The incorporation of UBS enhances the model's performance by improving its capacity to concentrate on informative and uncertain samples, thereby improving its decision boundaries.

The architecture for CRNet is mentioned in Table 13. Two parallel input branches process the data independently. A 16-unit gated layer with a *Tanh* activation function is used in the first branch, as shown in Table 13 (line 2). Two fully connected dense layers with 32 and 16 units are then used, both of which use the *Tanh* activation mentioned in Table 13 (line 3, line 4). Starting with a 1D convolutional layer with 64 filters, a kernel size of 5, and a *Sigmoid* activation function, the second branch uses a convolutional approach as illustrated in Table 13 (line 6). This is followed by an average pooling layer with a pool size of 2, as mentioned in Table 13 (line 7). After applying a second convolutional layer with the same kernel and filter sizes but a *Tanh* activation, an additional average pooling layer is applied, as shown in Table 13 (line 8, line 9). The output is then flattened and run through two dense layers with 120 and 84 units that are fully connected, both of which use *Tanh* activation. Prior to going through a final dense layer with a single neuron and a *sigmoid* activation function for binary classification, the outputs from the two branches are then concatenated.

Validation of proposed models on heart failure prediction dataset

To evaluate the generalizability of our proposed models, we tested them on the heart failure prediction dataset [36], which is a combination of five benchmark datasets including Cleveland with 303 observations, Hungarian with 294 observations, Switzerland with 123 observations, Long Beach VA with 200 observations, and the Stalog dataset with 270 observations. The combined dataset contains 918 instances with 12 features, where 508 correspond to healthy individuals and 410 to heart disease patients. Among the features, categorical variables such as *Sex*, *ChestPainType*, *RestingECG*, *ExerciseAngina*, and *ST_Slope* are originally of object type; therefore, label encoding is applied to convert them into integer format suitable for model training. Since the dataset is imbalanced, the OUPS balancing technique is applied to ensure fair representation of both classes and to reduce bias towards the majority class. After balancing, the dataset is split into 80% training and 20% testing sets. Furthermore, standard scaling is applied to normalize the feature space and bring all variables to a comparable scale, which improves the performance and convergence of the models.

The experimental results are listed in Table 14. Among the baseline models, GRU and LSTM performed competitively with accuracies of 78, precision values of 81 and 80, recall values of 78 and 75, and F1-scores of 81 and 80, respectively. These results show that recurrent units are capable of capturing temporal dependencies, though their recall values indicate some limitations in detecting positive cases consistently. LeNet achieved an accuracy of 78 with precision of 77, recall of 76, and F1-score of 81, highlighting its ability to extract spatial features but also showing limited sequential learning capacity. DNN achieved 77 accuracy with balanced precision and recall, but its log loss of 0.509 and relatively low ROC-AUC of 79 reflect that it does not capture temporal or spatial dependencies effectively. The ESN model produced the weakest results among baselines with 71 accuracy, 82 precision, 71 recall, 70 F1-score, and the highest execution time of 18 seconds, as its fixed reservoir weights restrict adaptability to complex feature distributions.

The proposed CRNet improved performance with 80 accuracy, 86 precision, 81 recall, 82 F1-score, and ROC-AUC of 89, along with a lower log loss of 0.465 compared to baseline and benchmark models. This demonstrates the advantage of combining convolutional layers with gated recurrent units to capture both spatial and sequential dependencies. Optimized CRNet further enhanced results with 82 accuracy, 80 precision, 91 recall, and 85 F1-score, supported by a log loss of 0.403. These outcomes validate the role of hyperparameter optimization through GOA in boosting recall and overall predictive robustness. Finally, CRNet with UBS outperformed all models by achieving 85 accuracy, 83 precision, 93 recall,

Table 14 Performance comparison of proposed models on heart failure prediction dataset

Metric	GRU	LSTM	LeNet	DNN	ESN	CRNet	Optimized CRNet	CRNet with UBS
Accuracy (%)	78	78	78	77	71	80	82	85
Precision (%)	81	80	77	76	82	86	80	83
Recall (%)	78	75	76	79	71	81	91	93
F1-Score (%)	81	80	81	77	70	82	85	88
ROC-AUC (%)	85	87	84	79	73	89	89	90
MCC (%)	56	58	57	55	50	61	64	71
Cohen's Kappa (%)	56	57	57	55	44	61	63	70
Log Loss	0.467	0.610	0.505	0.509	0.570	0.465	0.403	0.374
Execution Time (s)	7	10	5	5	18	12	18	25

Table 15 Performance comparison of proposed models with SMOTE and ADASYN balancing techniques

Metric	SMOTE			ADASYN		
	CRNet	Optimized CRNet	CRNet with UBS	CRNet	Optimized CRNet	CRNet with UBS
Accuracy (%)	87	87	91	87	89	90
Precision (%)	88	87	91	86	93	94
Recall (%)	87	87	91	85	84	87
F1-Score (%)	88	88	91	87	88	90
ROC-AUC (%)	95	94	96	95	96	96
MCC (%)	75	74	82	75	79	81
Cohen's Kappa (%)	75	74	82	75	78	81
Log Loss	0.261	0.263	0.193	0.267	0.230	0.204
Execution Time (s)	777.53	1281.66	1208.32	617.16	1328.23	792.82

88 F1-score, and the lowest log loss of 0.374, showing that CRNet with UBS significantly strengthens generalization by training on the most informative instances, even though it required the longest execution time of 25 seconds. Overall, the results confirm that the proposed 3-tier framework is generalizable. The improvements observed across CRNet, optimized CRNet, and CRNet with UBS demonstrate that each stage of the framework contributes meaningfully to robustness, efficiency, and predictive accuracy, establishing its reliability for broader medical applications such as heart failure prediction.

Comparative analysis with different balancing techniques

The results in Table 12 highlight that proposed models combined with OUPS balancing consistently achieve superior performance when compared with the SMOTE [37] and Adaptive Synthetic Sampling (ADASYN) [38] based results presented in Table 15. For example, CRNet with UBS using OUPS attains an accuracy of 94%, precision of 98%, recall of 90%, F1-score of 99%, and ROC-AUC of 99%. In contrast, the same model with SMOTE records only 91% accuracy, 91% precision, 91% recall, 91% F1-score, and 96% ROC-AUC, while with ADASYN it achieves 90% accuracy, 94% precision, 87% recall, 90% F1-score, and 96% ROC-AUC. The substantial improvement in recall with OUPS indicates that the method is much more effective at correctly identifying positive cases of heart disease, thereby reducing FNs which are critical in healthcare applications.

The optimized CRNet with OUPS further demonstrates this advantage, reaching an accuracy of 92%, precision of 93%, recall of 91%, and MCC of 82%. When compared to the optimized CRNet with SMOTE, which achieves only 87% accuracy, 87% precision, 87% recall, and 74% MCC, and with ADASYN, which reaches 89% accuracy, 93% precision, 84% recall, and 79% MCC, the gains with OUPS are clear. The higher MCC and Cohen's Kappa values with OUPS illustrate its ability to provide better overall balance in classification by improving the agreement between predicted and actual classes without biasing the model toward the majority class.

The proposed CRNet benefits from OUPS, showing 90% accuracy, 90% precision, 90% recall, 90% F1-score, and 97% ROC-AUC, compared to 87% accuracy, 88% precision, 87% recall, and 88% F1-score with SMOTE and 87% accuracy, 86% precision, 85% recall, and 87% F1-score with ADASYN. These results confirm that while SMOTE and ADASYN are effective in generating synthetic samples, they often introduce noisy or overlapping points. SMOTE relies on linear interpolation between minority samples, which can distort class boundaries, whereas ADASYN oversamples difficult instances aggressively, which may bias the model and increase the chance of overfitting.

SMOTE often creates synthetic points by simple interpolation, which can distort class boundaries, while ADASYN sometimes generates too many difficult samples, adding noise and increasing the risk of overfitting. In contrast, OUPS generates synthetic data by considering the underlying data distribution through propensity scores, which helps maintain the natural structure of the dataset. This allows the model to identify minority class cases more effectively and improves overall performance, especially in recall and MCC, which are critical for reliable heart disease prediction.

Comparative analysis with different active learning strategies

Entropy-Based Active Learning (EBAL) does not perform as well as UBS because it selects samples based on overall entropy of the predicted probabilities. While this strategy captures uncertain points, it often emphasizes noisy or less informative samples, which can negatively affect learning. As shown in Table 16, CRNet with EBAL achieves 90% accuracy, 94% precision, 87% recall, 90% F1-score, and 96% ROC-AUC. Although precision is reasonably high, the recall remains lower than UBS (90%), and the F1-score is significantly weaker than the 99% achieved by UBS. In addition, EBAL suffers from the highest log loss value of 0.204 and the longest execution time of 792 seconds, indicating less confident probability predictions and greater computational cost.

Table 16 Performance comparison of CRNet with different active learning strategies

Metrics	CRNet + EBAL	CRNet + MBAL	CRNet + UBS
Accuracy (%)	90	91	94
Precision (%)	94	92	98
Recall (%)	87	86	90
F1-Score (%)	90	92	99
ROC-AUC (%)	96	97	99
MCC (%)	81	84	88
Cohen's Kappa (%)	81	84	84
Log Loss	0.204	0.184	0.149
Execution Time (s)	792	98	385

These drawbacks highlight that EBAL is less effective in selecting truly informative samples for heart disease prediction.

Marginal-Based Active Learning (MBAL) also performs below UBS because it relies only on the difference between the top two predicted class probabilities to measure uncertainty. This approach sometimes overlooks cases where the model is broadly uncertain across all classes, which reduces its ability to capture highly informative samples. According to Table 16, CRNet with MBAL reaches 91% accuracy, 92% precision, 86% recall, 92% F1-score, and 97% ROC-AUC. While the F1-score and ROC-AUC are acceptable, recall is lower than UBS (86% versus 90%), showing that MBAL misses more TP cases. Its log loss of 0.184 is also higher than UBS (0.149), which indicates less confident predictions. The only advantage of MBAL is its efficiency, with execution time of just 98 seconds, but this comes at the cost of lower predictive strength compared to UBS.

In contrast, UBS achieves 94% accuracy, 98% precision, 90% recall, 99% F1-score, 99% ROC-AUC, an MCC of 88, Cohen's Kappa of 84, and the lowest log loss of 0.149. These results confirm that UBS selects the most truly informative samples near the decision boundary, leading to better balance between precision and recall, stronger probability calibration, and higher overall predictive power compared to EBAL and MBAL.

Overall, the results clearly demonstrate the superiority of UBS compared to EBAL and MBAL. The main advantage of UBS lies in its ability to focus on samples that lie near the decision boundary, which ensures a better trade-off between precision and recall, improved probability calibration, and overall stronger predictive performance. In contrast, EBAL suffers from the drawback of often prioritizing noisy or less informative samples, which leads to lower recall, weaker F1-score, and higher computational cost. Similarly, MBAL, although computationally efficient, overlooks global uncertainty by considering only the difference between the top two predicted classes. This limitation results in reduced recall and less confident probability estimates. Hence, while EBAL and MBAL present notable weaknesses in either informativeness or computational trade-offs, UBS effectively captures the most valuable samples, making it the most reliable active learning strategy for heart disease prediction in this study.

Comparative analysis with different optimization techniques

The comparison of CRNet with different optimization techniques are given in Table 17. CRNet with Bayesian Optimization (BO) [39] achieves accuracy 75%, precision 76%, recall 74%, F1-Score 75%, ROC-AUC 83%, MCC 50%, Cohen Kappa 50%, log loss 0.521 and execution time 1843 second. CRNet with Grid Search (GS) [40] achieves accuracy 85%, precision 80%, recall 93%, F1-Score 86%, ROC-AUC 94%, MCC 71%, Cohen Kappa 69%, log loss 0.321 and execution time 790 second. CRNet with GOA (Optimized CRNet) achieves accuracy 92%, precision 93%, recall 91%, F1-Score 92%, ROC-AUC 98%, MCC

Table 17 Performance comparison of CRNet with different optimization strategies

Metric	CRNet with BO	CRNet with GS	CRNet with GOA
Accuracy (%)	75	85	92
Precision (%)	76	80	93
Recall (%)	74	93	91
F1-Score (%)	75	86	92
ROC-AUC (%)	83	94	98
MCC (%)	50	71	82
Cohen Kappa (%)	50	69	84
Log Loss	0.521	0.321	0.194
Execution Time (s)	1843	790	763

82%, Cohen Kappa 84%, log loss 0.194 and execution time 763 second.

The poor results for BO in this run can be explained by several practical factors. Tuning deep networks is a noisy, expensive process and BO relies on a surrogate model built from those noisy evaluations; if the validation scores vary substantially between runs the surrogate can be misled and converge to a suboptimal region. BO also has overhead from sequential surrogate updates and acquisition optimization, so when the search space is high dimensional or contains mixed discrete and continuous choices its sampling budget can be effectively too small. The very large execution time of 1843 second indicates many expensive sequential evaluations or heavy surrogate overhead without enough useful exploration, which together explain the low accuracy, modest ROC-AUC 83% and the high log loss 0.521 that signals poorly calibrated probability estimates.

GS gives better recall and ROC-AUC than BO in this case because brute force sampling can hit parameter combinations that favor sensitivity, which explains the high recall 93% for CRNet with GS. However GS is inherently coarse and inefficient: it explores a fixed lattice and can miss fine-grained combinations that improve calibration and overall balance between precision and recall. That explains why GS has intermediate log loss 0.321 and precision 80% despite the high recall. Its execution time 790 second is lower than BO here because GS evaluations are likely parallelized or fewer in number, but the exhaustive nature still makes it inefficient relative to smarter search strategies.

GOA yields the best overall results because it is a population based metaheuristic that maintains diverse candidate solutions and balances exploration and exploitation. Population updates make it robust to noisy objective evaluations and allow discovering multiple promising regions of the hyperparameter space rather than getting trapped in a single local optimum. As a result GOA finds configurations that both improve discrimination and probability calibration, reflected in ROC-AUC 98% and the lowest log loss 0.194, while keeping execution time competitive at 763 second. The combination of better generalization (accuracy 92%, F1-Score 92%) and good calibration explains why CRNet tuned with GOA outperforms BO and GS on the metrics in Table 17.

Overall, the comparison clearly demonstrates that Optimized CRNet with GOA provides a more effective and balanced approach to hyperparameter tuning, offering strong generalization and well-calibrated probability estimates, while also maintaining efficiency in execution. In contrast, BO struggles with the noise and complexity of DL search spaces, leading to suboptimal convergence and excessive computational overhead. GS, although capable of achieving reasonable performance, suffers from coarse exploration and inefficiency, which limit its ability to capture fine-grained parameter interactions. These findings highlight that GOA not only outperforms traditional optimization strategies but also ensures robustness, efficiency, and clinical reliability of the CRNet framework.

K-Fold cross validation

The statistical method known as 10-FCV divides the dataset into ten subsets, or folds, in order to assess the performance and generalization of a model. One fold is reserved as a validation set and the other nine as training. Each step is repeated ten times, and the average of results over all folds is taken to provide a consolidated performance assessment that mitigates the bias of a single train-test split [41].

The working of 10-FCV is shown in Fig. 11. Results for the existing models, CRNet, Optimized CRNet, and CRNet with UBS using 10-FCV are presented in Table 12, alongside comparisons with existing models. The performance of the initial single evaluation, without applying cross-validation, is represented by the first bold value in each metric column of these models. The results for each fold are explained in detail by the mid-values, which help contextualize the variability and robustness of the model on various subsets of data. The final bold value represents the average performance across ten folds, summarizing the model's generalization ability. CRNet, Optimized CRNet, and CRNet with UBS achieve an average accuracy of 90, 92, and 94%. These models slightly have longer execution times within folds. Owing to their sophisticated architecture, these models outperform existing models in predictive accuracy and consistency; the computational cost is justified. Analyzing these results highlights the robustness of CRNet, Optimized CRNet, and CRNet with UBS and the effectiveness in generalizing across different subsets, even when compared to the existing models.

One-way ANOVA statistical test

A statistical test called One-Way ANOVA is used to assess whether the means of several groups differ significantly from one another. The way it operates is by comparing the variance within each group with the variance between groups [42]. This approach's main objective is to determine whether at least one group differs noticeably from the others. One-Way ANOVA is a great option for comparing the performance of various DL models because it can be used with multiple groups at once, unlike a t-test, which only compares two groups. The null hypothesis (H_0) in One-Way ANOVA states that any observed differences are the result of random variation and that all models perform equally on a particular metric. At least one group performs noticeably

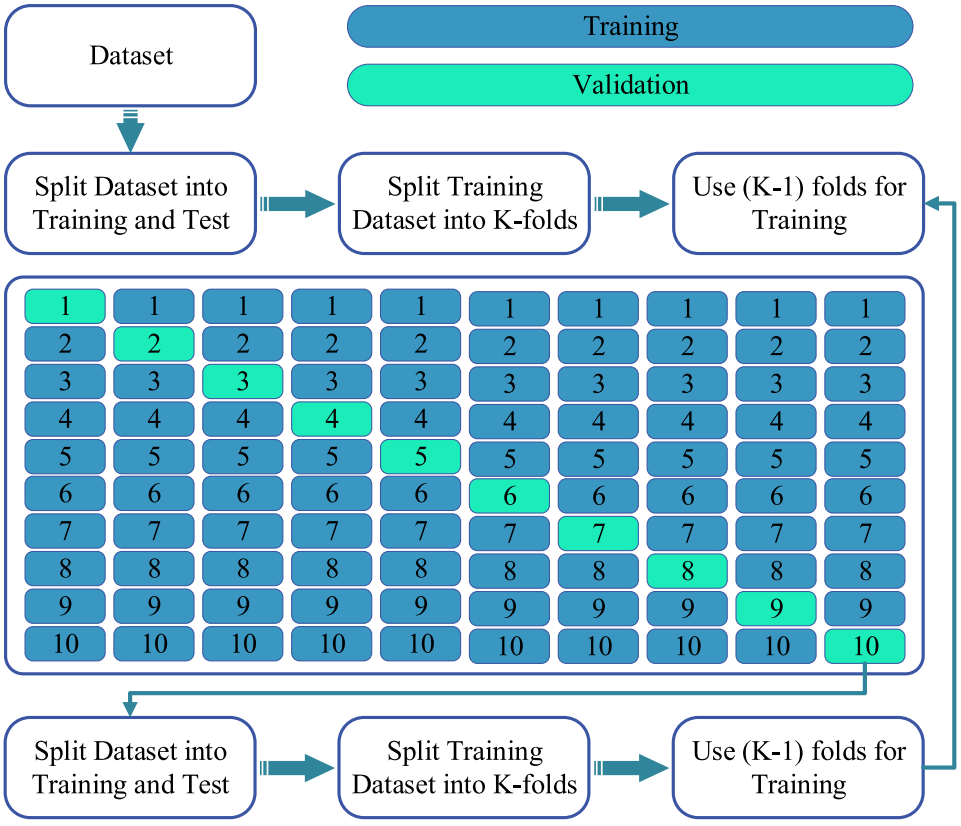


Fig. 11 Working of 10-fold cross validation

Table 18 One-way ANOVA results comparing proposed models with baseline models across performance metrics

Comparison	Metric	F-value	p-value
CRNet vs Baseline	Accuracy	47.192	0.00000
	Precision	68.353	0.00000
	Recall	189.796	0.00000
	F1-Score	171.000	0.00000
	ROC-AUC	23.516	0.00000
Optimized CRNet vs CRNet	Accuracy	20.147	0.00028
	Precision	46.800	0.00000
	Recall	5.832	0.02660
	F1-Score	5.828	0.02665
	ROC-AUC	0.379	0.54562
CRNet with UBS vs CRNet	Accuracy	44.882	0.00000
	Precision	171.204	0.00000
	Recall	0.101	0.75415
	F1-Score	236.391	0.00000
	ROC-AUC	44.596	0.00000

different according to the alternative hypothesis (H_a). The F-statistic and the p-value are two important statistical values that are produced by the test. The F-statistic is a ratio that quantifies the degree of the variation between groups relative to variation within each group. According to the p-value, a higher F-statistic means that there are more statistically significant differences between the groups. To confirm the significant difference between the groups, the null hypothesis is rejected if the p-value falls below a predetermined threshold, usually 0.05.

Table 18 highlights the statistical comparison of proposed models with baseline models using one-way ANOVA across multiple evaluation metrics. The comparison of CRNet with baseline models shows that all metrics achieve extremely high F-values with corresponding p-values of 0.00000, such as accuracy at 47.192,

precision at 68.353, recall at 189.796, F1-score at 171.000, and ROC-AUC at 23.516, which confirms that the proposed CRNet significantly outperforms traditional baseline approaches. When comparing optimized CRNet with standard CRNet, the analysis indicates meaningful improvements in accuracy with an F-value of 20.147 and a p-value of 0.00028, precision with an F-value of 46.800 and a p-value of 0.00000, as well as recall and F1-score with F-values of 5.832 and 5.828, both significant at p-values close to 0.026. Although ROC-AUC shows an F-value of 0.379 with a p-value of 0.54562, suggesting no statistically significant difference, the optimized CRNet still achieves better generalization in most metrics. Finally, the comparison of CRNet with UBS against proposed CRNet demonstrates highly significant improvements in accuracy, precision, F1-score, and ROC-AUC with F-values of 44.882, 171.204, 236.391, and 44.596 respectively, all with p-values of 0.00000, while recall shows an insignificant difference with an F-value of 0.101 and a p-value of 0.75415. These results establish the significance of the proposed enhancements, where optimization improves the robustness of CRNet, and the integration of UBS further refines its predictive capability, especially in scenarios with limited labeled data.

Local interpretable model-agnostic explanations

LIME is a technique designed to explain predictions of ML models, preferably of complex models such as DL or ensemble methods, which are often considered black boxes. LIME tries to explain model predictions locally rather than interpreting the whole model [43].

In LIME, we approximate a complex model with a simpler, interpretable model (typically a linear model) near a prediction. First, a sample prediction is selected, and perturbations of the data are generated, i.e., we slightly modify the data to what would be its neighbors. As a next step, it is used to predict outcomes for the perturbed samples. A weight is provided to each of the modified samples, signifying its degree of similarity or proximity to the original data point. Then, the complex model is trained on a perturbed data set, along with the predictions, to create a simpler model, like LR [44]. In the local region, to mimic whether the complicated model functions in that small area, this simpler model serves as a local surrogate. LIME analyzes the coefficients of the simpler model and finally generates an explanation by analyzing the features contribution to the prediction. It can understand individual predictions. Simple models like LR allow us to have an easy interpretation of feature importance. Specifically, LIME helps to provide insight into choices behind the model's prediction by demonstrating that particular features affect the prognosis. In addition, LIME is an agnostic interpretability tool in the sense that it is not bound to a specific type of ML model, i.e., it doesn't require a DT, an ensemble method, or a DL model; it can simply handle all types of ML and DL models.

In the case of the CRNet model, Fig. 12 illustrates the effects of the model's predictions for heart disease given specific feature values. LIME approximates the complex

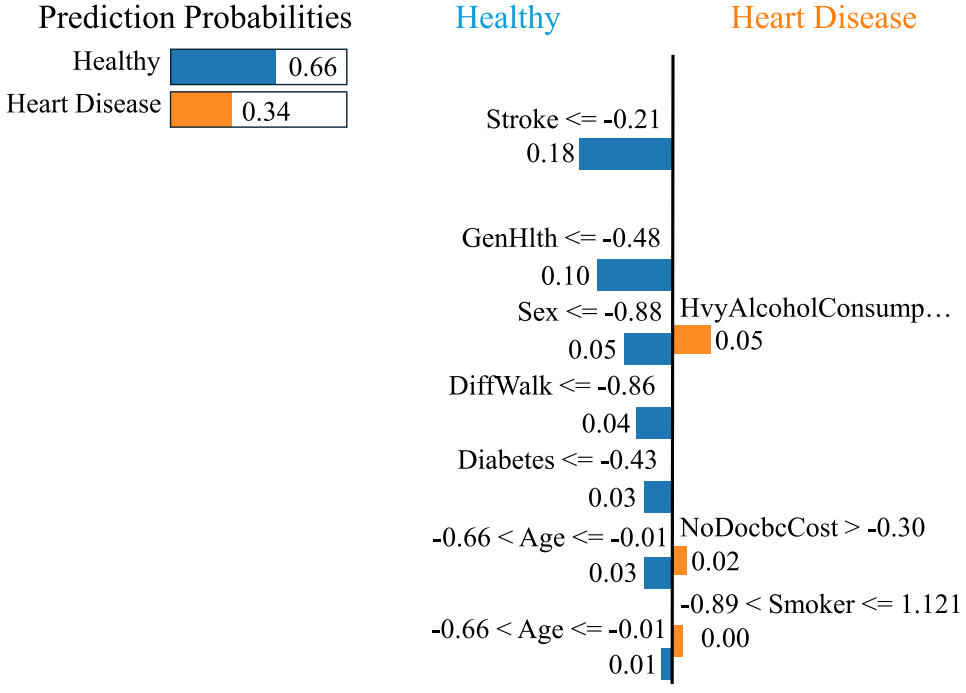


Fig. 12 LIME prediction for CRNet for accurate heart disease prediction at early stages

CRNet model behavior in the vicinity of each prediction with an interpretable model whose behavior is explainable. In Fig. 12, features like *Diabetes*, *MentHlth*, *PhysActivity*, *Age*, and *Sex* influence the predictions made by the model. Features such as *Stroke* produce positive contributions for heart disease and *Physical_Activity* negative contributions for lowering the risk. The color-coded representation is intuitively shown, featuring values and their influence, provides essential information on the factors influencing CRNet predictions.

The model assigns a probability of 0.98 for the patient being *Healthy* and 0.02 for *Heart Disease*. Key protective indicators include absence of *Stroke* $<= -0.21$, good *GenHlth* $<= -0.48$, female *Sex* $<= -0.88$, no *DiffWalk* $<= -0.45$, normal *HighChol* $<= -0.86$, absence of *HighBP* $<= -0.87$, no *Diabetes* $<= -0.43$, younger *Age* $-0.66 to 0.01$, and positive *Fruits* consumption $-1.32 to 0.76$. These features push the prediction strongly towards the healthy class. In contrast, small contributions from *HvyAlcoholConsump* (0.05), *NoDocbcCost* > -0.30, and *Smoker* $-0.39 to 1.12$ increase risk slightly but remain overshadowed by stronger protective values. Clinically, this reflects that absence of major risk factors such as hypertension, diabetes, cholesterol, and stroke history, combined with healthy lifestyle behaviors, significantly lowers heart disease risk despite minor negative influences.

The LIME summary plot for the CRNet shown in Fig. 13 shows the overall contribution of the most important features in predicting heart disease. Negative contributions such as *Stroke, GenHlth, Sex, HvyAlcoholConsump, DiffWalk, HighChol, HighBP,* and *Diabetes* indicate that higher values of these risk factors reduce the likelihood of a negative prediction and instead strengthen the model's decision towards heart disease presence. Similarly, conditions on *Age, Smoker,* and *Fruits* also show varying impacts, where younger age ranges and lower fruit intake contribute towards heart disease risk. Features like *NoDocbcCost* represent access to healthcare, further influencing the prediction. Overall, the plot provides an interpretable understanding of how clinical and lifestyle factors collectively drive the model's classification of heart disease.

The probability of Optimized CRNet in Fig. 14 shows that the model predicts a 95% chance of heart disease and only 5% chance of being healthy. On the right, the features with their values explain this decision. *Age* in the range $0.18 < \text{Age} \leq 0.71$ increases risk, as older age is strongly linked to cardiovascular disease. *Stroke* > -0.37 contributes positively, reflecting the higher chance of heart problems after a stroke. *DiffWalk* > 1.01 also raises the prediction, since reduced mobility is often associated with poor cardiovascular health. Smoking in the range $0.31 < \text{Smoker} \leq 1.00$ further increases risk by damaging blood vessels. *Diabetes* > 0.61 adds to the prediction, as it is a major contributor to heart disease. In contrast, low cholesterol (HighChol ≤ -1.21) pushes slightly toward the healthy outcome, showing a protective effect. Overall, the model combines these clinical factors to explain why the patient is more likely to have heart disease.

The LIME summary plot for the Optimized CRNet shown in Fig 15 shows the overall contribution of the most important features in predicting heart disease. *Age,*

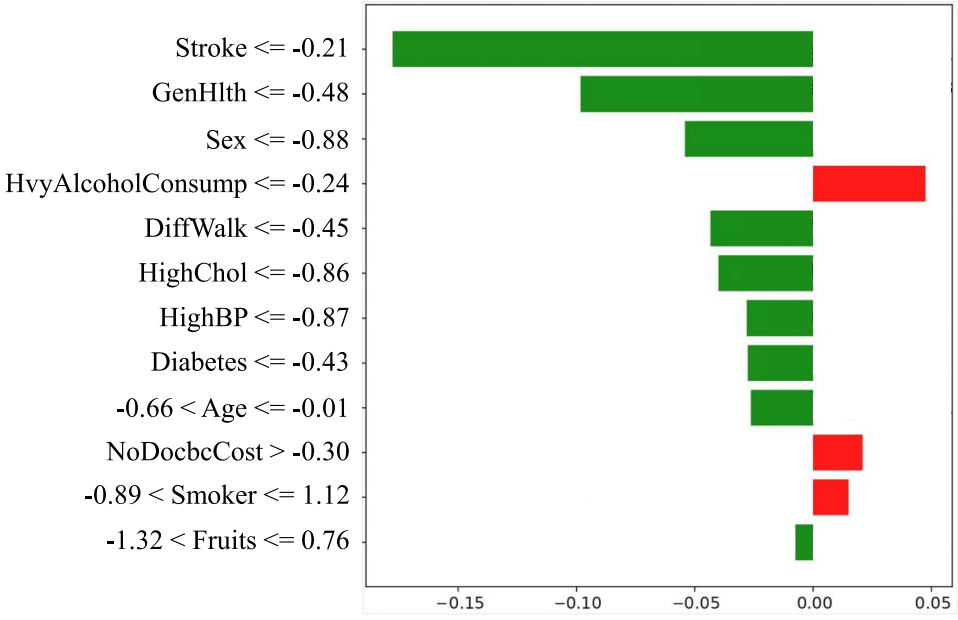


Fig. 13 LIME summary for CRNet for accurate heart disease prediction at early stages

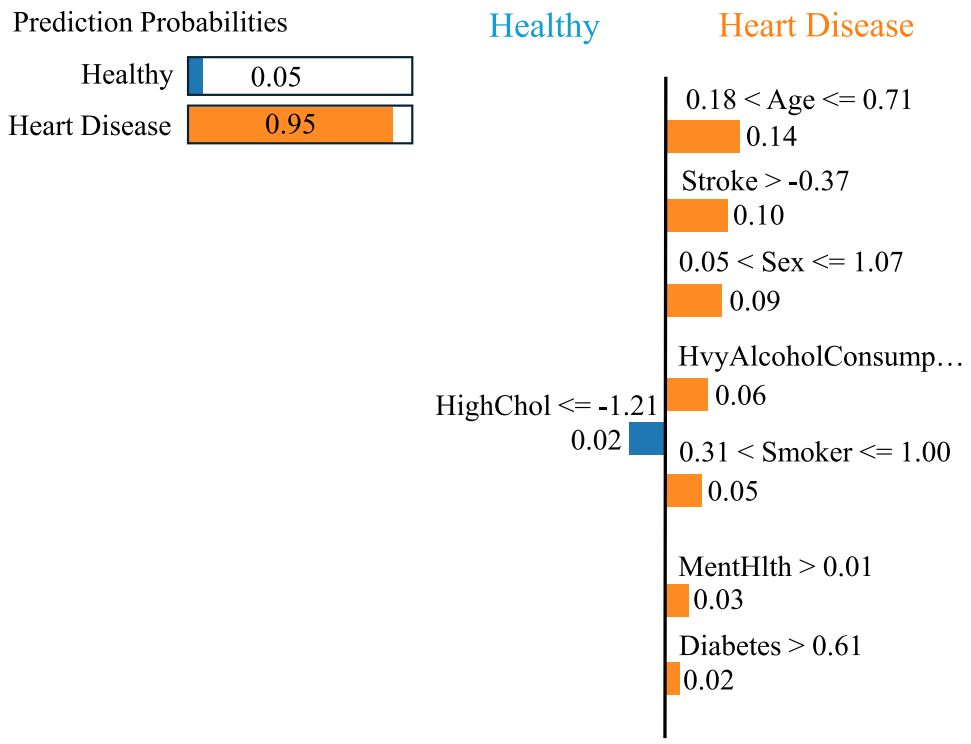


Fig. 14 LIME prediction for optimized CRNet for accurate heart disease prediction at early stages

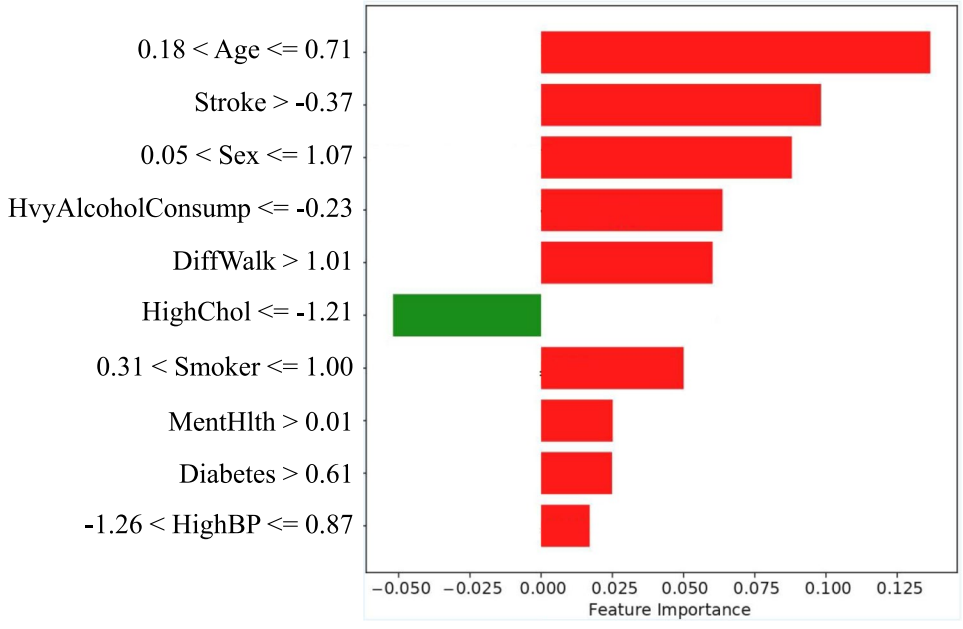


Fig. 15 LIME summary for optimized CRNet for accurate heart disease prediction at early stages

Stroke, DiffWalk, Smoker, Diabetes, and HighBP consistently push predictions toward the disease class, reflecting their strong clinical relevance as major cardiovascular risk factors. On the other hand, lower values of *HighChol* contribute in the opposite direction, slightly supporting the healthy outcome. This indicates that the model mainly relies on well-established risk factors such as *Age, Stroke, Diabetes, Sex, Smoker,* and mobility limitations when predicting heart disease, while also recognizing that low cholesterol levels can reduce the likelihood of disease.

The probability Figure illustrated in 16 shows that the model predicts a 99% chance of heart disease and only 1% chance of being healthy, indicating a very strong

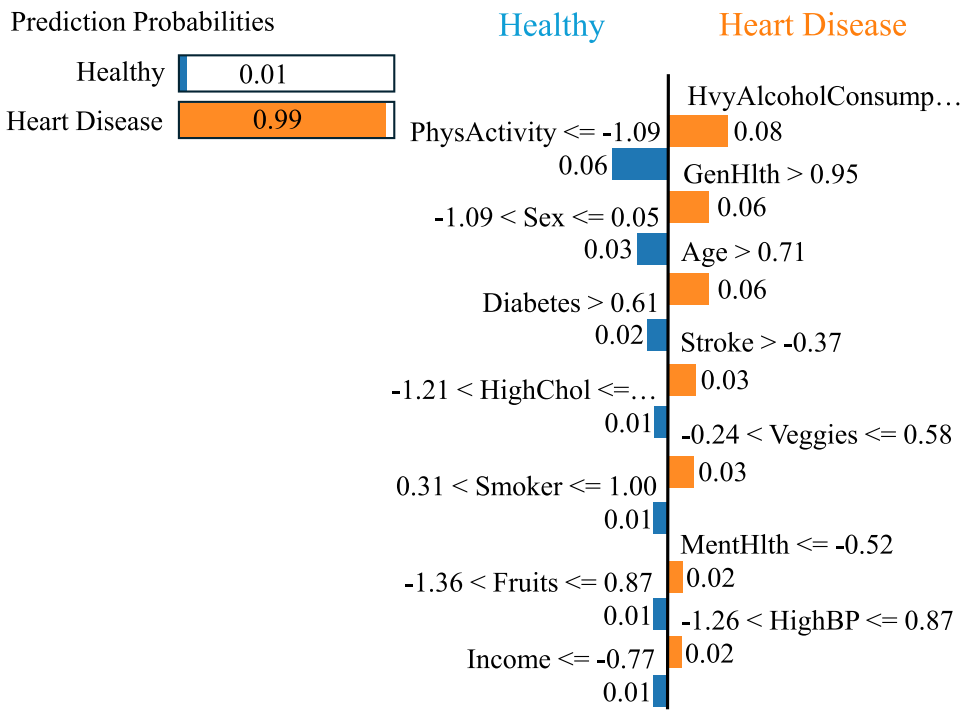


Fig. 16 LIME prediction for CRNet for UBS for accurate heart disease prediction at early stages

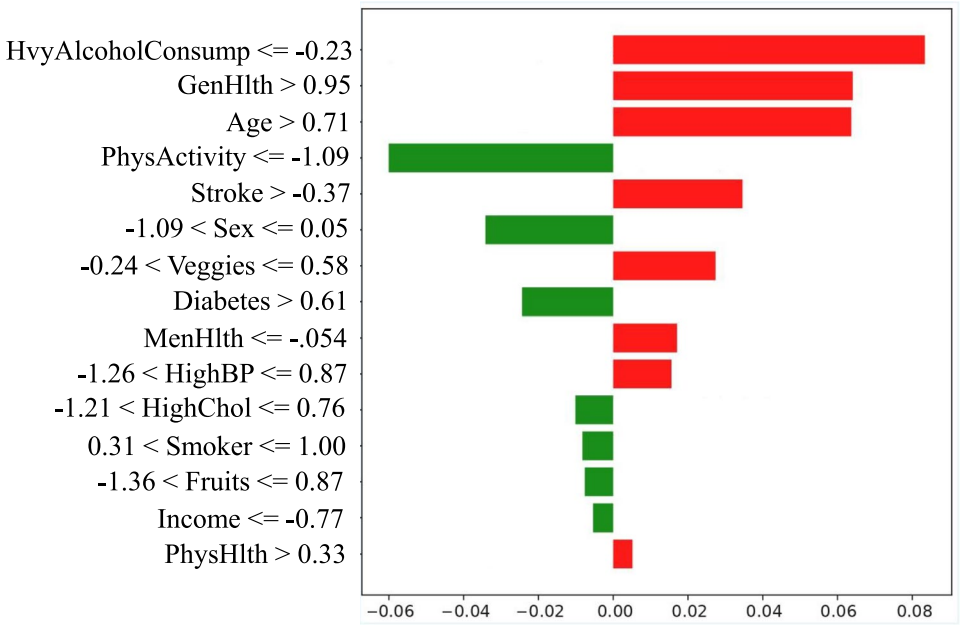


Fig. 17 LIME summary for CRNet with UBS for accurate heart disease prediction at early stages

prediction toward disease. On the right, the feature contributions explain this outcome. *HvAlcoholConsumption* contributes positively to the prediction, as excessive alcohol use is linked to cardiovascular problems. *GenHlth* >0.95 and *Age* >0.71 both strongly increase the risk, reflecting poor perceived health and the effect of older *Age* as major risk factors. *Stroke* >-0.37 also adds to the prediction, consistent with the high cardiovascular risk in patients with prior stroke. *Veggies* ≤ -0.24 indicate low intake, which supports the disease outcome since poor diet is a known contributor. Similarly, *MentHlth* ≤ -0.52 and *HighBP* >0.87 both push the prediction toward heart disease, as hypertension and psychological stress are associated with adverse outcomes. In contrast, some factors such as *PhysicalActivity* ≤ -1.09, higher *Fruits* intake ≤ -1.36, lower *HighChol* ≤ -1.21, and lower

Income ≤ -0.77 contribute slightly toward the healthy class, but their influence is outweighed by the stronger disease-related features. Overall, the model explains the 99% prediction as driven mainly by age, stroke, high blood pressure, poor general health, and lifestyle-related risk factors.

The LIME summary plot shown in Fig. 17 for CRNet with UBS shows that features such as *heavy alcohol consumption, poor general health, older age, prior stroke, low vegetable intake, poor mental health, diabetes, and high blood pressure* consistently push predictions toward heart disease, while protective factors like *higher fruit intake, lower cholesterol, and physical activity* slightly favor the healthy outcome. Overall, the model mainly relies on well-established cardiovascular risk factors, with lifestyle and clinical conditions such as *alcohol* use, *hypertension*, *diabetes*, and *stroke* emerging as the strongest contributors to predicting heart disease. LIME is essential for improving the interpretability of intricate models such as CRNet, Optimized CRNet, and CRNet with UBS in the healthcare industry so that medical practitioners can comprehend and have faith in the models' forecasts. Ultimately, LIME helps create better and more reliable healthcare results by connecting medical knowledge with AI decision-making through clear explanations of how different factors contribute.

Shapley additive explanations

SHAP is an explainable AI technique that enables ML and DL models to explain their predictions by assigning the feature importance score based on the impact they have on the model prediction. It is based on cooperative game theory, utilizing Shapley values to determine the average contribution across all possible feature

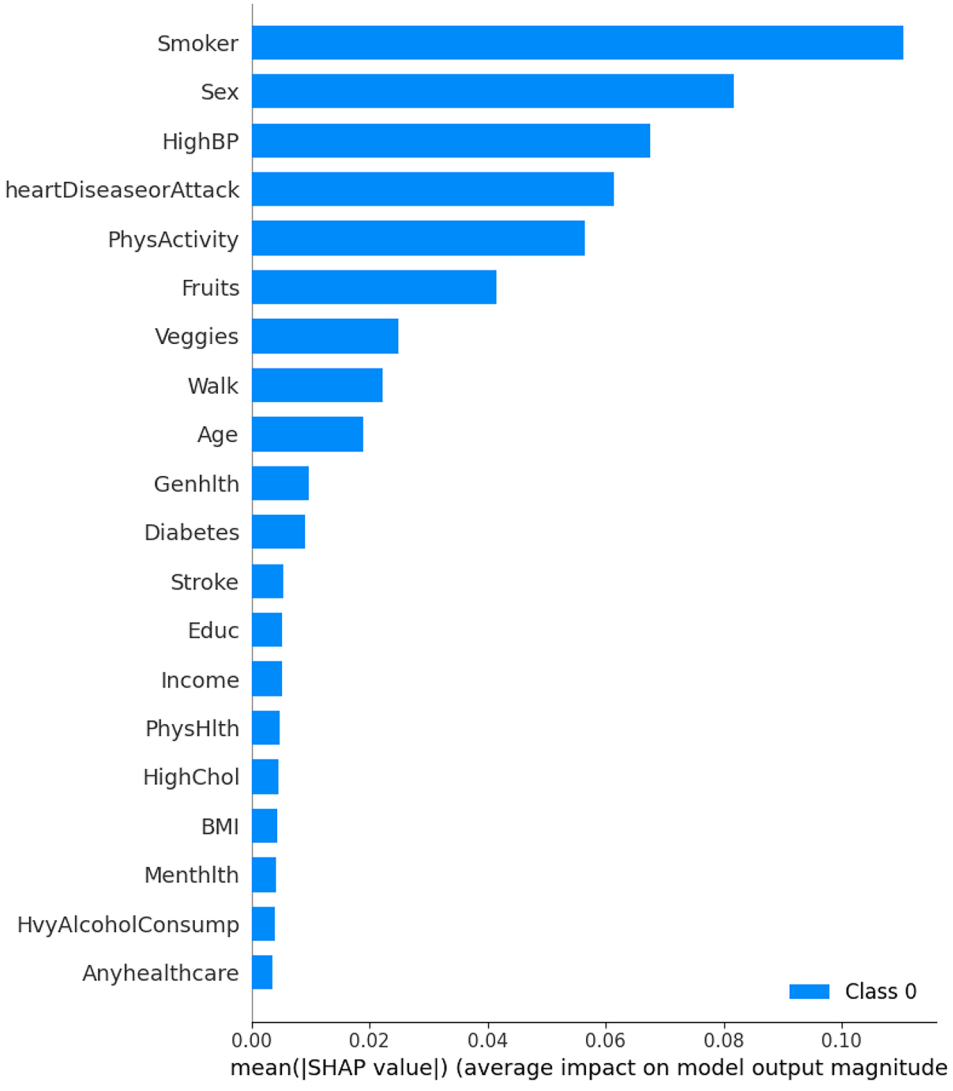


Fig. 18 SHAP summary for CRNet for accurate heart disease prediction at early stages

combinations [45]. SHAP explains that a value between −1 and 1 is assigned to each feature, showing the role that every feature has played in the model's predictions. To assist stakeholders in understanding what constitutes a prediction, this technique is often employed to promote model transparency, determine which factors could be making the models uncommon, and build trust in AI systems with interpretable results.

Figure 18 illustrates the SHAP summary plot. It shows the importance of features in the CRNet model in the prediction of heart disease. Features such as *Smoker*, *Sex*, *HighBP*, and *heartDiseaseorAttack* show more contribution in predicting heart disease. Figure 18 represent the SHAP value for a feature in a given prediction, colored by the feature values. However, we can see the effect of each feature value on the model's predictions: *smoker* has the largest impact, though in a positive way for the predictions.

Figure 19 is a violin plot that shows the distribution of SHAP values of key features and their influence on heart disease risk. Features such as *income*, which generally has positive SHAP values and therefore has a reduced risk of heart disease, or *HighBP* and *Sex*, which have negative SHAP values and have a higher risk of heart disease. The violin width at different SHAP values is the density of instances and shows the consistency of these feature contributions over the dataset.

A detailed breakdown of the waterfall plot prediction for one instance is given in Fig. 20. It exhibits the process of adding or subtracting the SHAP values of particular

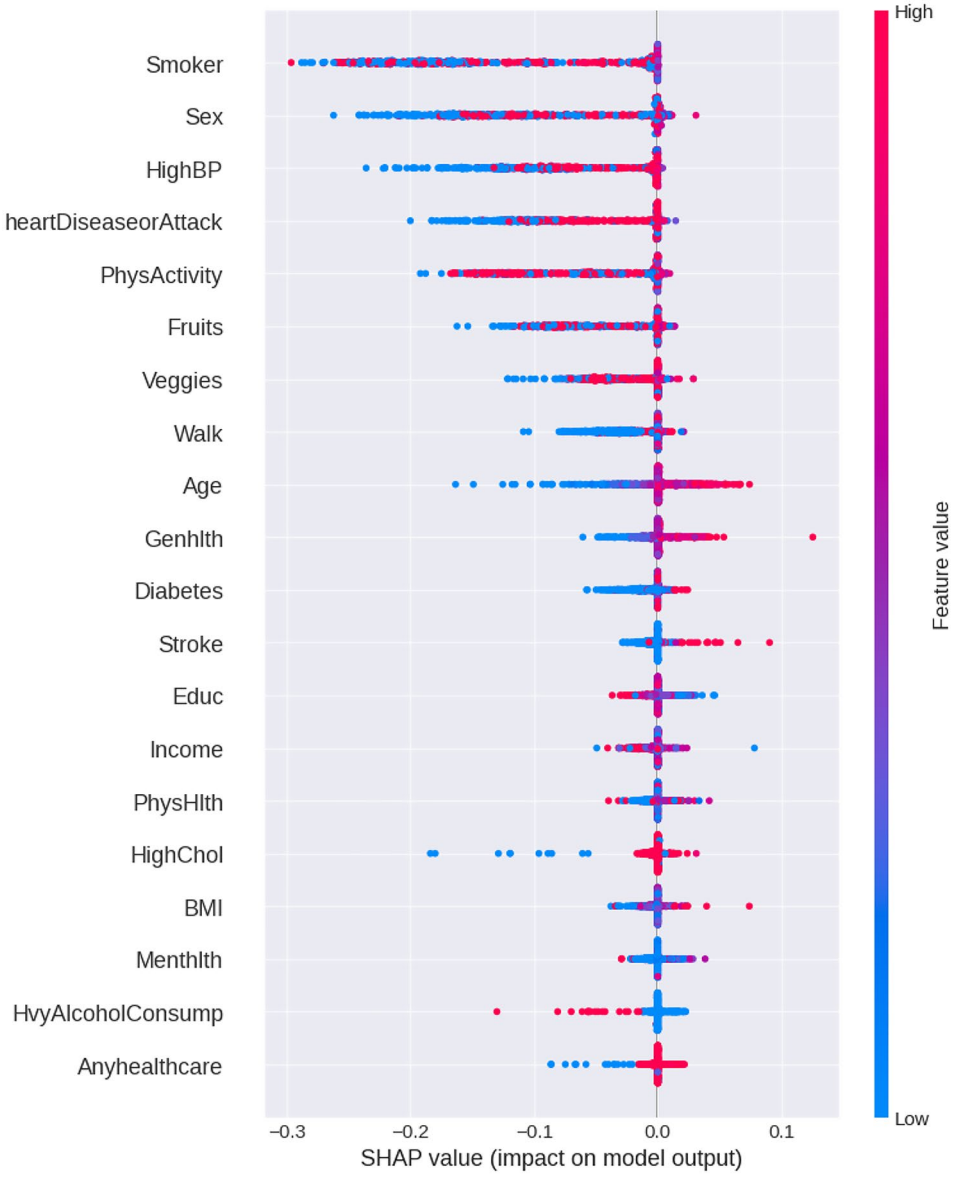


Fig. 19 Voilin plot for CRNet for accurate heart disease prediction at early stages

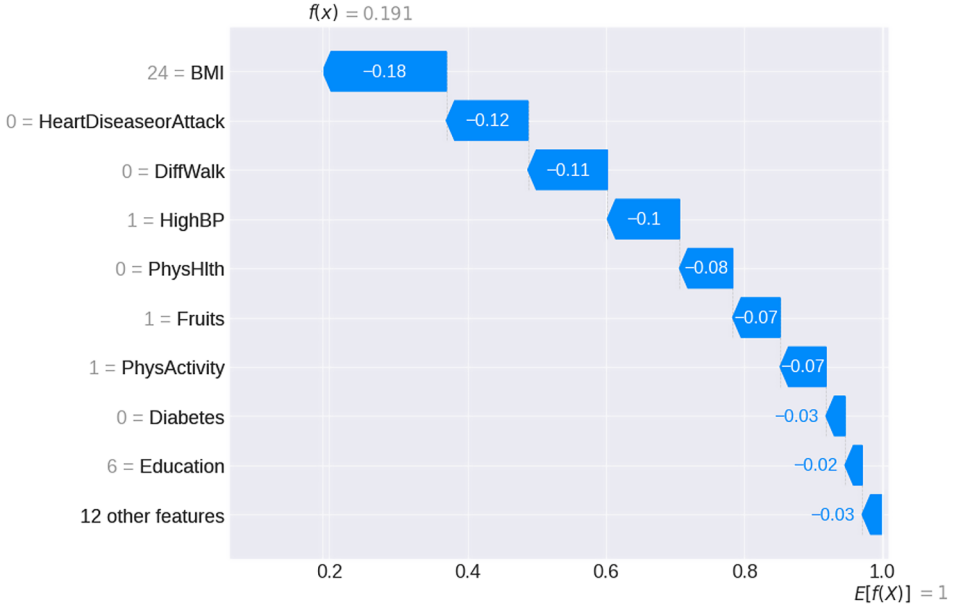


Fig. 20 Waterfall plot for CRNet for accurate heart disease prediction at early stages

feature values to determine the total contributions of those features from a baseline value. The model incurs a negative SHAP value of –0.18 with the feature *BMI*, making the prediction more probable in the healthy class, and a negative SHAP value of –0.1 with the feature high *blood_pressure* makes the odds of heart disease higher.

The violin plot in Fig. 21 illustrates the distribution of the most influential features contributing to the optimized CRNet model. The feature *GenHlth* shows a wide and skewed distribution, highlighting its strong variability and dominant role in prediction. The feature *Age* presents a moderately spread distribution, reflecting its consistent yet less variable impact compared to *GenHlth*. The categorical feature *Sex* displays two distinct density regions, indicating differences in predictive influence between male and female groups. Both *HighBP* and *HighChol* exhibit denser distributions toward higher values, suggesting that elevated blood pressure and cholesterol strongly contribute to the classification outcomes. Finally, *Smoker* demonstrates a clear separation in distribution, reinforcing its significant effect in distinguishing between risk groups. Overall, the violin plot emphasizes that *GenHlth*, followed by *Age*, *Sex*, *HighBP*, *Smoker*, and *HighChol*, are the leading predictors in the optimized CRNet framework.

The waterfall plot in Fig. 22 highlights the individual feature contributions to the optimized CRNet prediction. The feature *Age* with a negative contribution of -2.175 decreases the predicted risk, while *HvyAlcoholConsump* contributes positively with the largest impact of 4.876, strongly increasing the likelihood of heart disease. Similarly, *Sex* (1.066) and *Smoker* (0.999) both push the prediction upward, indicating that being male and smoking behavior elevate the risk. In contrast, *HighChol* shows a negative contribution of -1.211, slightly lowering the overall prediction score. Collectively, these contributions demonstrate how both lifestyle and demographic factors interact to shape the final decision of the optimized CRNet model.

The violin plot in Fig. 23 presents the top contributing features identified by CRNet with UBS, where *Age* and *GenHlth* emerge as the most dominant predictors, reflecting the strong influence of age groups and self-reported general health in heart disease risk. The feature *Sex* shows clear density separation, emphasizing gender-based differences in predictions. Both *HighBP* and *HighChol* exhibit dense upper distributions, signifying their critical role in elevating risk. The lifestyle factor *Smoker* further strengthens classification impact with distinct density patterns, while *Diabetes* contributes significantly, aligning with its established role as a major cardiovascular risk factor. Together, these features demonstrate that CRNet with UBS effectively prioritizes clinically relevant indicators, capturing both medical history and lifestyle-related determinants.

The waterfall plot in Fig. 24 illustrates the feature contributions to the CRNet with UBS prediction. The feature *Age* (0.902), *HighBP* (0.875), *Sex* (1.066), and *HighChol* (0.925) all show positive contributions, collectively increasing the predicted likelihood of heart disease. In contrast, *Fruits* contributes negatively with -0.341, slightly reducing the risk score. This balance highlights how demographic, medical, and lifestyle factors interact to shape the model's final decision.

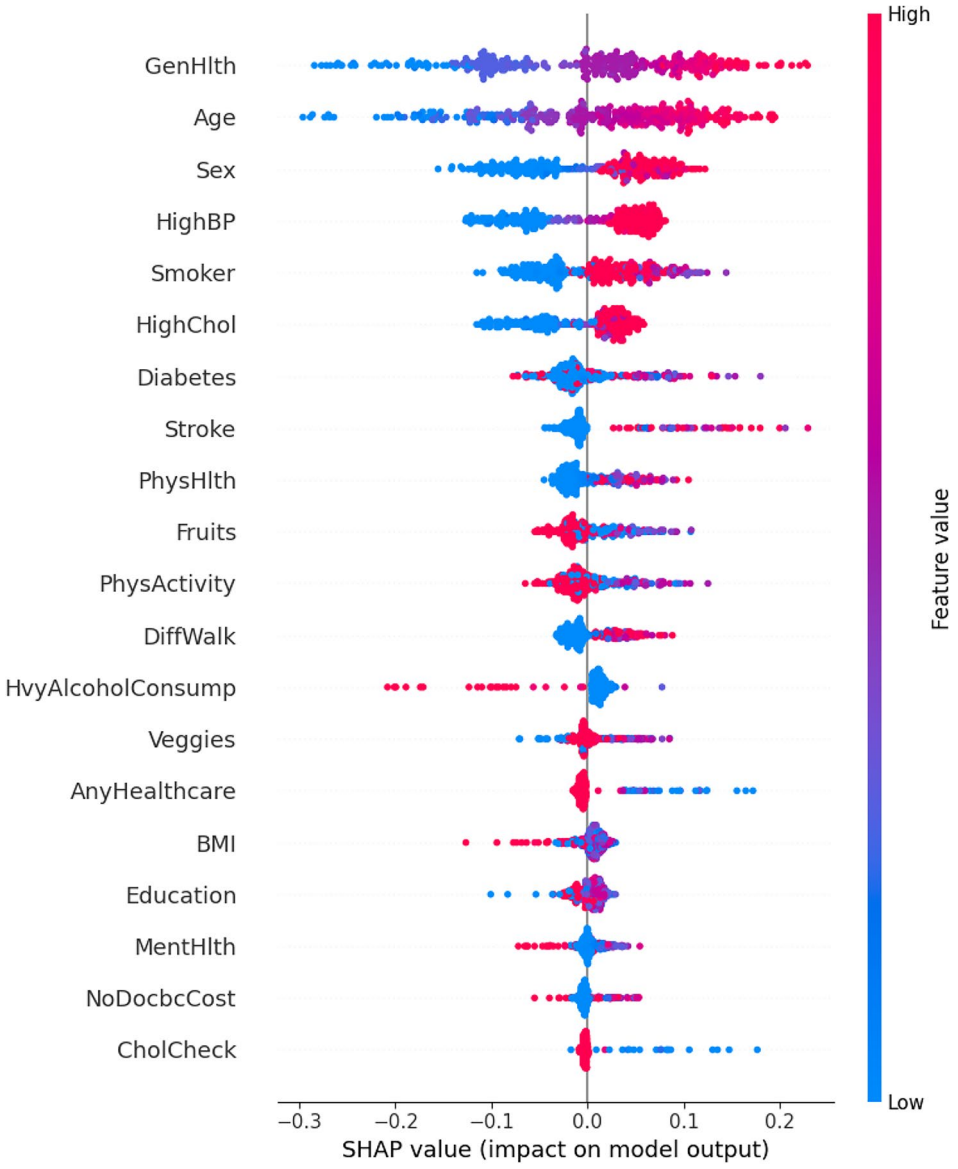


Fig. 21 Voilin plot for optimized CRNet for accurate heart disease prediction at early stages

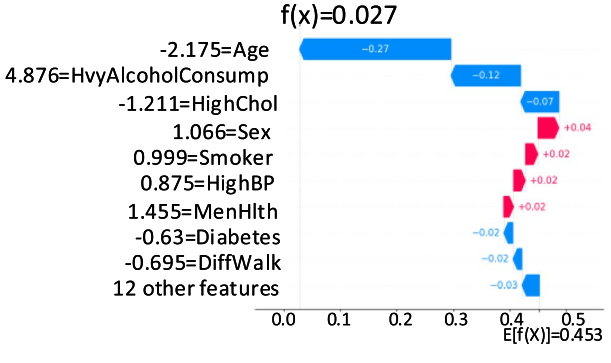


Fig. 22 Waterfall plot for optimized CRNet for accurate heart disease prediction at early stages

SHAP analysis makes the proposed models easier to understand by showing how different features affect the prediction of heart disease. Stakeholders can learn more about the model's decision-making process by analyzing feature impacts through summary, violin, and waterfall plots. This degree of transparency promotes trust in AI-driven healthcare systems because their predictions align with medical logic and empirical observations.

Comparative analysis of local interpretable model-agnostic explanations and SHapley additive explanations

LIME and SHAP are two of the most well-known approaches for interpreting DL models in the field of explainable AI. Although these approaches aim to improve the interpretability of complex models, they

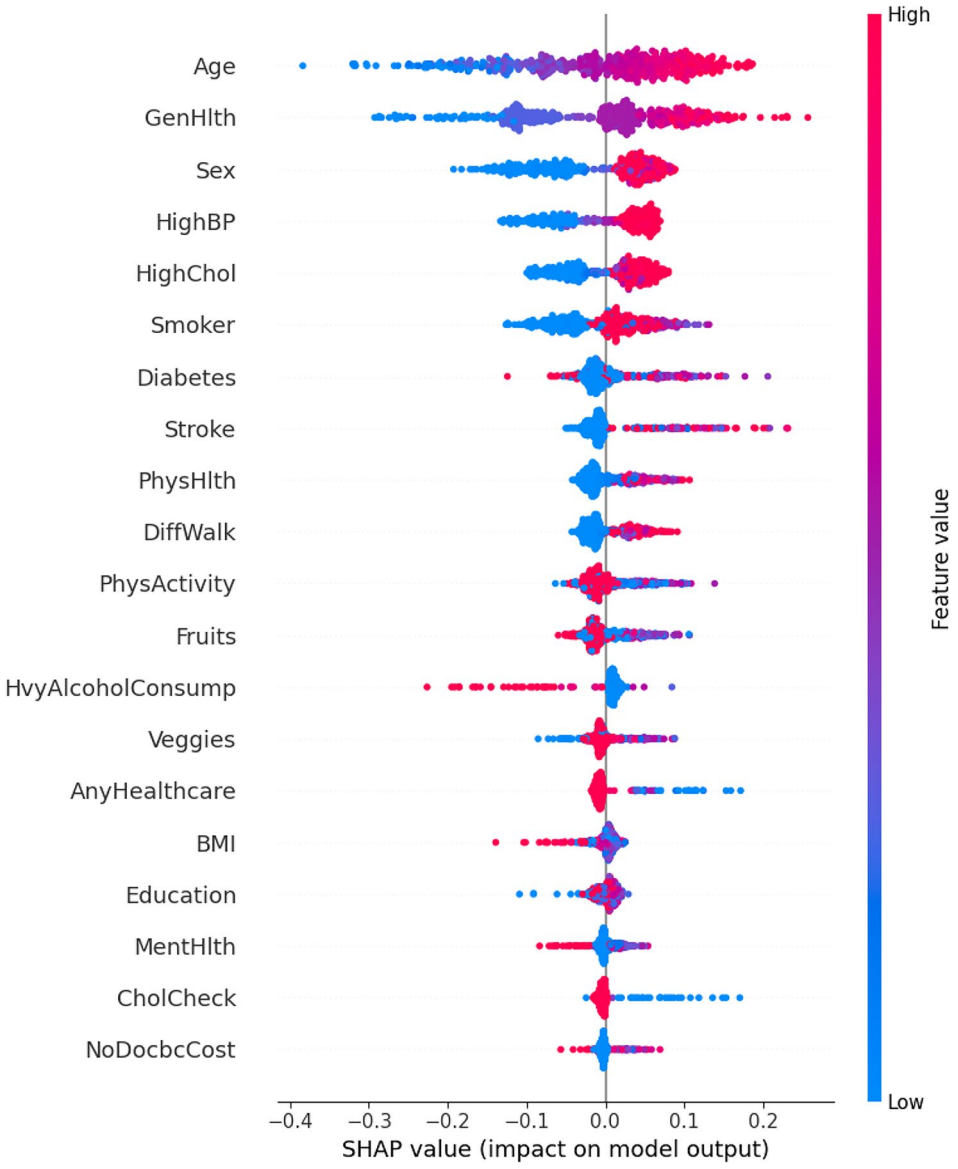


Fig. 23 Voilin plot for CRNet with UBS for accurate heart disease prediction at early stages

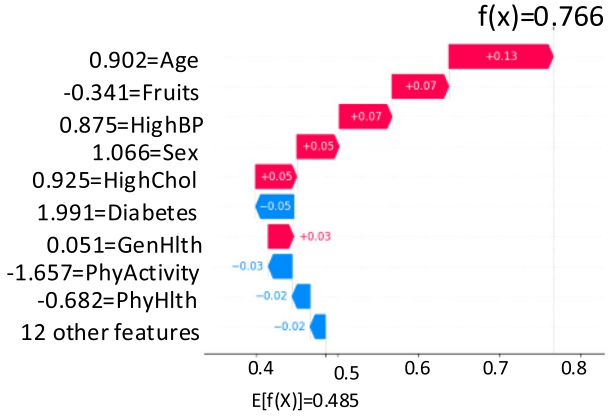


Fig. 24 Waterfall plot for optimized CRNet with UBS for accurate heart disease prediction at early stages

strongly differ in terms of methodology, computational demands, and the breadth of insights they offer. SHAP offers explanations based on Shapley values, a well-known idea for distributing gains among coalition members fairly. It allocates each feature's contribution to a prediction by considering every possible subset of features, ensuring that its explanations are globally consistent and additive. The theoretical rigor of SHAP guarantees its mathematical soundness and adherence to fairness and symmetry principles. LIME adopts a more heuristic approach, locally approximating the complex model using an interpretable surrogate model such as LR. It fits the surrogate model to simulate the local decision boundary, creates perturbed samples around the instance

being explained, and monitors the black-box model's predictions.

LIME is effective at capturing local interpretability, but it doesn't have the theoretical assurances of global consistency that SHAP does. The computational requirement is one of the primary differences between SHAP and LIME. Especially for models with a large number of features, SHAPs reliance on determining, Shapley values, which entail assessing every potential feature subset, can become computationally costly. While methods such as KernelSHAP make an effort to approximate these values, deep models or large datasets may still require a lengthy process. The interpretability of the feature importance scores generated by SHAP and LIME differs. Through the integration of explanations from several predictions, SHAP provides a thorough understanding of feature significance. This helps users understand that each element impacts both specific predictions and the model overall. Moreover, the additive property of SHAP guarantees that the output of the model is the sum of the contributions of all features, providing a comprehensive and coherent explanation. The primary focus of LIME is local interpretability. It is skilled at elucidating specific predictions by emphasizing the characteristics most accountable for a given result. Nevertheless, it is less appropriate for comprehending the model's wider behavior because its explanations are restricted to the area around the data point under analysis. Furthermore, the sampling strategy employed for perturbations can affect LIMEs explanations, potentially resulting in discrepancies.

The explanations generated by both LIME and SHAP demonstrate a strong alignment with well-established clinical knowledge. Features such as *smoking, high blood pressure, diabetes, age,* and *physical activity* factors universally recognized in cardiology as key risk indicators for heart disease, are consistently highlighted as influential in the model's predictions. This concordance confirms that the model is not merely detecting statistical correlations but is capturing relationships that have direct clinical relevance. By reflecting the same risk factors that physicians consider in diagnosis and preventive strategies, the explanations strengthen confidence in the reliability and practical value of our framework. Furthermore, this alignment enhances the interpretability of the model for healthcare practitioners, bridging the gap between computational predictions and real-world clinical decision-making.

Conclusion and future work

Many different risk factors collectively contribute to heart disease. Early detection and accurate diagnosis are challenging due to complex and multifaceted symptoms. Late diagnosis leads to high mortality rates, emphasizing the need for effective predictive models. OUPS was employed to address the problem of data imbalance, ensuring that instances of minority classes were fairly represented while maintaining the natural distribution of the data. In this study, we proposed 3-TIFF for early heart disease prediction. The first tier consist of CRNet, which successfully combines gated units to capture temporal dependencies and convolutional layers for spatial feature extraction. In the second tier, to improve generalization and performance, we utilized the GOA to systematically optimize the hyperparameters of CRNet. In the third tier, the framework incorporated UBS, an active learning strategy that enables the model to rank the most informative samples for annotation. The results demonstrated that the proposed framework outperformed the existing models. CRNet, optimized CRNet, and CRNet with UBS achieved an accuracy score of 90, 92, and 94%, precision score of 90, 93, and 98%, recall score of 90, 91, and 90%, F1-score of of 90, 92, and 99%, ROC-AUC values of 97, 98, and 99%, MCC scores of 81, 82, and 88%, and Cohen's Kappa scores of 83, 84, and 88%, while reducing log loss to 0.210, 0.194, and 0.149, respectively. Furthermore, 10-FCV was used on existing models, CRNet, optimized CRNet, and CRNet with UBS to make sure they were reliable and applicable to a wide range of data splits, ensuring good performance. We further performed One-Way ANOVA to verify the statistical significance of results, ensuring that performance variations were not due to random fluctuations. Finally, we employed LIME and SHAP explainable AI methods to improve model interpretability and explainability.

Limitations

Although the proposed CRNet framework demonstrates strong predictive performance and interpretability, several limitations should be acknowledged. First, the interpretability tools have not yet been validated with domain experts, which limits confirmation of their clinical reliability. Second, while the dataset used in this study is publicly available and de-identified to minimize ethical, privacy, and legal concerns, the framework has not yet been tested on more sensitive patient data where stricter compliance may be required. Finally, the model and framework is developed and evaluated on a single dataset, which may restrict their generalizability to multi-center or real-world clinical environments.

Future work

In future work, we will focus on using CRNet on various datasets, including electronic health record, imaging, and real-world data, to increase its generalizability. We also plan to incorporate external validation datasets to further demonstrate the robustness and reliability of the proposed framework. We will also investigate more advanced DL architectures and optimization techniques

to achieve higher prediction accuracy. Furthermore, we plan to integrate privacy-preserving mechanisms to ensure the protection of patient data when employing state-of-the-art AI solutions in healthcare. These steps will enhance the robustness, accuracy, and trustworthiness of the proposed framework, making it more suitable for deployment in diverse clinical settings.

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Author contributions
Aymin Javed: Writing original draft, Methodology, Validation; Nadeem Javaid: Supervision, Review, Visualization; Muhammad Shafiq: Formal analysis, Conceptualization, Investigation; Jin-Ghoo Choi: Validation, Formal analysis, Methodology; Nabil Alrajeh: Funding, Review, Resources.

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Data availability
The datasets used in this study are available at:https://www.kaggle.com/code/alexteboul/heart-disease-health-indicators-datasetnotebook.https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction.

Declarations

Ethics approval and consent to participat
Not applicable.

Competing interests
Authors declare no competing interests.

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